Render, Don't Decode: Weight-Space World Models with Latent Structural Disentanglement

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Abstract

Training world models on vast quantities of unlabelled videos is a critical step toward fully autonomous intelligence. However, the prevailing paradigm of encoding raw pixels into opaque latent spaces and relying on heavy decoders for reconstruction leaves these models computationally expensive and uninterpretable. We address this problem by introducing **NOVA**, a world modelling framework that represents the system state as the weights and biases of an auxiliary coordinate-based implicit neural representation (INR). This structured representation is analytically rendered, which eliminates the decoder bottleneck while conferring compactness, portability, and zero-shot super-resolution. Furthermore, like most latent action models, NOVA can be distilled into a context-dependent video generator via an action-matching objective. Surprisingly, without resorting to auxiliary losses or adversarial objectives, NOVA can disentangle structural scene components such as background, foreground, and inter-frame motion, enabling users to edit either content or dynamics without compromising the other. We validate our framework on several challenging datasets, achieving strong controllable forecasting while operating on a single consumer GPU at $\sim$40M parameters. Ultimately, structured representations like INRs not only enhance our understanding of latent dynamics but also pave the way for immersive and customisable virtual experiences.

1 Introduction

The capacity to predict plausible future states within an environment is a cornerstone of physical intelligence. World models [Ha and Schmidhuber, 2018] achieve this by learning transition dynamics conditioned on actions, enabling embodied agents to plan and act through internal simulation (e.g. robotic manipulation and autonomous driving). Recently, the abundance of unlabelled, action-free online videos has driven the emergence of **latent action (world) models** [Garrido et al., 2026; Bruce et al., 2024; Schmidt and Jiang, 2023]. Devoid of ground-truth action signals during training, they must discover underlying control mechanisms purely from observation, bringing them closer to video generation models, which learn to produce coherent sequences of frames from noise or context.

The predominant paradigm for training world models and video generation models involves projecting and reconstructing frame sequences through a latent space, for example, using a video tokeniser [Bruce et al., 2024]. A prominent class of autoregressive models encode individual frames into *abstract* latent vectors via a variational autoencoder [Kingma et al., 2013], then map resulting predictions back to pixels via a learned spatial decoder. This paradigm has been immensely productive [Hafner et al., 2020; Micheli et al., 2022; Bruce et al., 2024], but carries intrinsic limitations. First, the decoder consumes more storage, adding a number of hardly interpretable parameters that could limit

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usability in safety-critical scenarios [Rudin, 2019]. As a case in point, Bruce et al. [2024, pp. 6, 23] find that as they scale, a much larger decoder is essential, accounting for up to 87% of their video tokeniser. Second, any change in output resolution requires expensive retraining or post-hoc upsampling workarounds [Chambon et al., 2025]. Third, the latents it consumes lack portability, as they are tethered to that specific decoder. Furthermore, they have no semantic or geometric interpretations. Current attempts to disentangle semantic components such as background, identity, and motion from an abstract latent space require auxiliary losses [Guen and Thome, 2020], adversarial objectives [Denton et al., 2017], or architectural decompositions [Hsieh et al., 2018], all of which add training complexity without structural guarantees.

Coordinate-based **implicit neural representations** (INRs) [Sitzmann et al., 2020; Dupont et al., 2022] represent an image signal not as a pixel array but as the weights of a neural network, thus encoding a continuous function $\mathbb{R}^2 \to \mathbb{R}^C$ that maps any spatial coordinate to its RGB ($C = 3$) or greyscale ($C = 1$) pixel value. They gained prominence by powering novel view synthesis frameworks such as NeRF [Mildenhall et al., 2021], and have recently demonstrated compression capabilities that surpass JPEG at low bitrates [Mostajeran et al., 2025]. Crucially, further advantages arise from an analytical rendering process that circumvents the need for a spatial decoder. Despite inherent spectral aliasing challenges [Barron et al., 2021; Lindell et al., 2022], properly constrained parameter-free rendering intrinsically supports continuous, arbitrary-resolution querying at test time, a capability that is critical in domains such as extreme weather events forecasting, where a single pixel can encapsulate a vast area of the globe (e.g., over 784 km^2 in state-of-the-art global AI models [Lam et al., 2023]); in such contexts, unlocking granular resolution can be the difference between life and death for massive populations. Furthermore, unlike abstract encodings, INRs are highly portable, as their analytical formulation makes them easily sharable across the scientific community. Given these fundamental strengths, *structured* representations like INRs and Gaussian splats [Kerbl et al., 2023] are increasingly recognised as the most promising technologies to power next-generation scientific modelling, visual perception, and computer graphics.

In this paper, we argue that INRs are a potent representation upon which world models and video generation models can be built. To realise this, we leverage the paradigm of **weight-space learning** (WSL), which views the weights and biases of neural networks such as INRs as data points for a higher-level learning system [Schürholt et al., 2024]. This paradigm was recently utilised by Li et al. [2025a] and Nzoyem et al. [2026] to achieve remarkable performance on time series modelling benchmarks. These WSL approaches, however, remain untested on video modalities. Crucially, the latter proved unstable, requiring weight, gradient, and value clipping to prevent numerical explosion. Any viable weight-space world model must first address these acute stability issues.

To bridge this gap, we introduce **NOVA** (**N**eural **O**ntology for **V**isual **A**bstraction), a stable and physically consistent world model framework that evolves dynamics entirely in the weight space of an implicit *neural* representation. Its straightforward architecture induces a three-level semantic *ontology* that mirrors human intuition about object-centric video structure: (i) the **background** ($\bar{\boldsymbol{z}}$) is captured by a single set of base weights shared across the entire dataset, learned to encode everything common to all videos; (ii) the **foreground content** ($\boldsymbol{z}_t$) is defined by per-frame weight offsets encoding what is present in the current video frame but not shared globally; and (iii) the **motion** ($\boldsymbol{u}_t$) is an inter-frame action signal that dictates how a latent state transitions to the next. To achieve such *visual abstraction*, our framework leverages an additive formulation that cleanly isolates these properties during both forward dynamics and base composition prior to rendering.

Specifically, our key contributions are as follows:

1. We build NOVA, the first world model defined by a weight-space representation. We demonstrate that the inclusion of a base representation $\bar{\boldsymbol{z}}$ addresses the characteristic instability of weight-space model training. Similar to recent literature [Schmidt and Jiang, 2023], we show that this framework can be tuned to produce actions autoregressively during inference, enabling seamless transition from world modelling to context-conditioned video generation.

2. We show that the $(\bar{\boldsymbol{z}}, \boldsymbol{z}_t, \boldsymbol{u}_t)$ parametrisation induces a semantically meaningful three-level decomposition, i.e. background, state, and motion, purely by construction with no auxiliary or adversarial losses. This in turn empowers users to "edit" the generation process, whereby importing frames or actions from an arbitrary sequence cleanly maps to new sequences, a process we call **zero-shot content (or motion) retargeting** (Figure 1). This hints that,

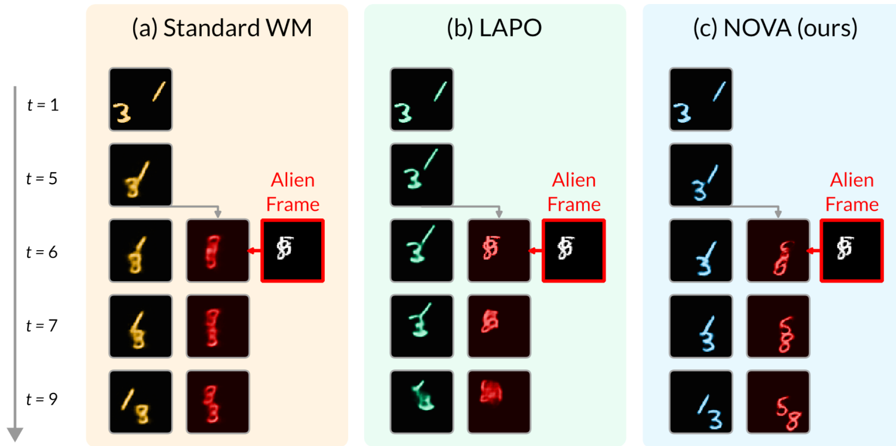

Figure 1: **Video editing facilitated by object-motion disentanglement**. Conditioned on the same initial frame, all models must generate future states; at $t = 6$, this state is abruptly replaced with an alien frame's encoding. **(a)** The standard world model (WM), trained in an abstract latent space with NOVA's decomposition strategy, manages to separate content from motion. **(b)** LAPO [Schmidt and Jiang, 2023] generates clean frames, but allows the foreign frame to completely alter its dynamics; furthermore, its generation quality degrades over extended horizons (starting at $t = 9$). **(c)** NOVA generates visually striking digits over long horizons and swaps their identities without altering the underlying dynamics (e.g., a 5 seamlessly inherits the trajectory of a 1).

contrary to expectations from recent work [Daniel et al., 2026], global action vectors can support local dynamics disentanglement when paired with a structured base representation.

3. We highlight the numerous benefits of INRs as structured latent representations for video and world models (e.g., parameter-efficiency, compactness, portability, and resolution-independence). Remarkably, NOVA maintains **zero-shot super-resolution** capabilities even when querying coordinate grids that are significantly denser than those used during training. To facilitate this, we implement a simple but principled post-training Nyquist-Shannon frequency masking procedure that provably prevents aliasing at arbitrary scale factors.
4. We provide a comprehensive set of benchmarks to showcase the practical usefulness of our method in sequential forecasting. We show this with experiments across three carefully selected video datasets (Moving MNIST, PhyWorld, and WeatherBench) covering continuous physics, rigid-body collisions, and geospatial dynamics, all trained on a single consumer GPU at roughly 40M parameters. Furthermore, these datasets serve as the basis for ablation studies that probe the NOVA architecture and reveal its inner workings.

While our approach yields several practical benefits, we emphasise that our primary contribution lies in demonstrating that world models can be constructed entirely from structured weight-space representations rather than raw pixels. In doing so, we propose an effective disentanglement technique applicable to standard world models, as observed in Figure 1. To elucidate this contribution, Section 2 outlines the experimental setup, followed by a detailed exposition of our method in Section 3. We present and analyse main results in Section 4, and conclude with a discussion of the method's broader implications and limitations in Section 5.

2 Experimental Setup

Problem Setting. We consider an environment generating a sequence of 2D observations $\mathbf{o}_t \in \mathbb{R}^{H \times W \times C}$ for $t \in \{1, \ldots, T\}$, with $T \geq 2$. World modelling[2] aims to discover transition dynamics $\boldsymbol{z}_{t+1} \approx \mathcal{T}(\boldsymbol{z}_t, \boldsymbol{u}_t)$, where $\boldsymbol{z}_t \in \mathbb{R}^{d_z}$ denotes a (latent) representation of the underlying state, from which the observation $\mathbf{o}_t$ can be reconstructed; $\boldsymbol{u}_t \in \mathbb{R}^{d_u}$ denotes actions typically known and part of the training data. When ground-truth actions are unavailable, the model is self-supervised and must infer $\boldsymbol{u}_t$ from $\boldsymbol{o}_t$ and $\boldsymbol{o}_{t+1}$; we refer to such models as *latent action models* (LAMs).

[2] To eliminate confusion from inconsistent definitions, we adopt the term "world model" to refer solely to the transition model, excluding adjacent components that encode states or infer actions, similar to [Schmidt and Jiang, 2023].

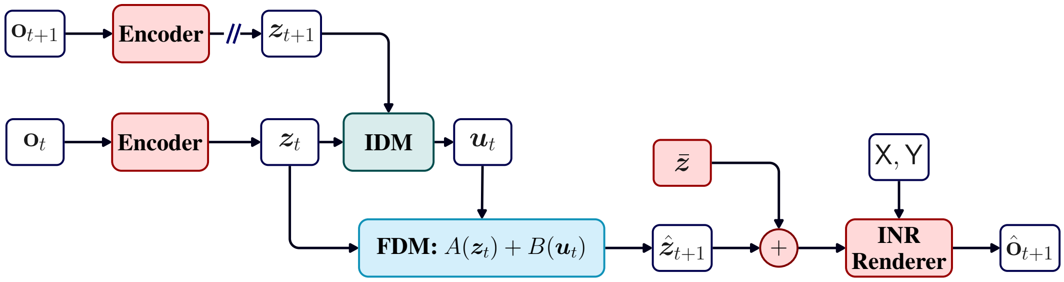

Figure 2: **NOVA architecture.** The shared Encoder (red) maps frames to weight-space offsets z_t. During training, the IDM (green) infers latent action u_t from consecutive encoded states. The FDM (blue) predicts the next weight offset via the additive mapping $A(z_t) + B(u_t)$. Unlike conventional decoders, the Renderer (red) is not trained; it is an analytical function that maps a coordinate grid (X, Y) to pixel values via an implicit neural representation (INR). The sign // denotes a stop-gradient sg operator, included to reduce memory overheads during training.[3]

Because both system states and actions may live in a "latent" space, throughout the paper, we prioritise the term *weight space*, as defined in [Zhou et al., 2023], for the space in which the dynamics evolve, i.e., $\mathbb{R}^{d_z}$, and *action space* for the actions themselves, i.e., $\mathbb{R}^{d_u}$.

Datasets. We train and evaluate on three main datasets spanning diverse dynamics: (1) **Moving MNIST** [Srivastava et al., 2015] is a seminal multi-entity dataset showing two digits moving at constant velocity and bouncing on a grid. Velocity direction and magnitude are randomly selected, and the kinematics are deterministic; digits overlap without collision, and bouncing reflections are non-linear. (2) **PhyWorld Collision 30K** [Kang et al., 2025] comprises roughly 30 thousand simulated rigid-body collisions with varying ball radii and velocities. It tests OOD generalisation and strict physical consistency (momentum and kinetic energy conservation). (3) **WeatherBench (2m temperature)** [Rasp et al., 2020] is a processed version of the ERA5 archive [Hersbach et al., 2020], containing global atmospheric data recorded at hourly intervals from 1979 to 2018. Specifically, we utilise the 2-meter temperature subset at a 5.625° resolution, challenging models to predict atmospheric flows that exhibit highly complex and non-linear behaviour.

Baselines. We benchmark our approach against two representative paradigms. First, a **standard WM** (see Figure 8 in Section A) is a LAM that utilises an encoder-decoder setup for fair comparison against NOVA. This baseline shares NOVA's training configurations, and differs only in its use of an abstract latent space (of comparable size) rather than the weight space to progress its dynamics. Second, we compare against **LAPO** [Schmidt and Jiang, 2023], a framework that removes the encoder-decoder setup by operating directly in pixel space. LAPO infers discrete action codes using an inverse dynamics network with two VQ codebooks [Van Den Oord et al., 2017] optimised via EMA. Additionally, we include **WARP** [Nzoyem et al., 2026] as a baseline in Section D.5.

Metrics. To quantify temporal consistency and structural preservation, we evaluate predictions against the empirical ground truth observations and their corresponding distribution of pixel intensities. We assess identity drift and distributional overlap using the **Wasserstein-1** (W_1) distance, **Jensen–Shannon Divergence** (JSD), and **Bhattacharyya** distance (B). Spatial and frequency-domain structural integrity are measured via the **Structural Similarity Index Measure** (SSIM) and **FFT magnitude distance** ($d_{\mathcal{F}}$), respectively. Furthermore, to avoid direct pixel-level comparisons, **Mean Squared Error** (MSE) is applied to evaluate the accuracy of object coordinates, masses, and velocities. Additional experimental datasets and formally defined metrics can be found in Section C.1.

3 The NOVA Framework

3.1 Architecture Overview

Figure 2 shows the NOVA architecture, comprising components we discuss in two categories.

[3] We note that the omission of the sg operator does not lead to trivial representational collapse [Bardes et al., 2023].

Encoding and Rendering. The Encoder is a strided convolutional backbone that maps each frame $\mathbf{o}_t \in \mathbb{R}^{H\times W\times C}$ to a compact vector $\boldsymbol{z}_t \in \mathbb{R}^{d_{\boldsymbol{z}}}$, where $d_{\boldsymbol{z}}$ is chosen to match the number of learnable parameters in the coordinate-based rendering network. We maintain a single set of base parameters $\bar{\boldsymbol{z}}$ learned across the entire training dataset, and pixels are rendered independently to reconstruct the future frame, in parallel:

$$\hat{\mathbf{o}}_{t+1} = \mathrm{MLP}_{\boldsymbol{\theta}_{t+1}}(\mathsf{x},\mathsf{y}),\ \forall(\mathsf{x},\mathsf{y}) \in \mathsf{X}\times\mathsf{Y}, \quad \text{where} \quad \boldsymbol{\theta}_{t+1} \triangleq \mathtt{unflat}(\bar{\boldsymbol{z}} + \boldsymbol{z}_{t+1}). \tag{1}$$

Coordinates are normalised with X and Y regularly sampled in the range $[-1, 1]$. The `unflat` operation turns a flat vector into MLP [McCulloch and Pitts, 1943] weight and bias tensors to be used alongside non-linear activation functions and Fourier feature encoding [Tancik et al., 2020].

Anchoring to $\bar{\boldsymbol{z}}$ provides two benefits: (i) it stabilises training by framing the prediction $\boldsymbol{z}_t$ as a weight-space residual, enabling the model to output small-magnitude offsets, analogous to residual networks [He et al., 2016]; and (ii) it provides a dataset-level prior that absorbs static background, freeing $\boldsymbol{z}_t$ to encode only dynamic content, akin to meta-learning initialisations [Finn et al., 2017].

Inverse and Forward Dynamics. The Inverse Dynamics Model (IDM) is an MLP that receives two consecutive encoded states $(\boldsymbol{z}_t, \boldsymbol{z}_{t+1})$, and outputs the latent action $\boldsymbol{u}_t$ explaining the transition. Rather than predicting $\boldsymbol{z}_{t+1}$ through a monolithic block, we employ an *additive* Forward Dynamics Model (FDM): $\boldsymbol{z}_{t+1} = A(\boldsymbol{z}_t) + B(\boldsymbol{u}_t)$, where $A : \mathbb{R}^{d_{\boldsymbol{z}}} \to \mathbb{R}^{d_{\boldsymbol{z}}}$ and $B : \mathbb{R}^{d_{\boldsymbol{u}}} \to \mathbb{R}^{d_{\boldsymbol{z}}}$ are independent MLPs. A operates on the current weight offset alone, while B maps the latent action into the weight space, encoding how the action modulates the INR.

This forward decomposition is inspired by linear state-space models [Gu et al., 2021; Nzoyem et al., 2026] and extends them to nonlinear settings. Another key difference is that A does not prescribe autonomous dynamics, but acts as a dedicated object detector [Terver et al., 2026]. Our A/B decoupling is the mechanism through which the content/motion disentanglement emerges.

3.2 Training Procedure

Training generally proceeds in three phases, each targeting a different component of the model.

Phase 1. We pre-train the encoder and base latent $\bar{\boldsymbol{z}}$ by minimising the independent per-frame reconstruction loss $\mathcal{L}_1 = \frac{1}{T}\sum_{t=1}^{T} \|\mathbf{o}_t - \hat{\mathbf{o}}_t\|_2^2 + \lambda \cdot \mathrm{SSIM}(\mathbf{o}_t, \hat{\mathbf{o}}_t)$, where $\hat{\mathbf{o}}_t$ is computed using Equation (1), and the λ-weighted SSIM term prevents excessive smoothing [Wang et al., 2004].

While this phase is optional, it remains essential, as it exposes the encoder to not just the initial frames, but to all T time steps of each sequence, thereby encouraging more robust feature extraction.

Phase 2. With the encoder and base latent $\bar{\boldsymbol{z}}$ frozen, we jointly train the FDM and IDM by minimising the latent transition objective:

$$\mathcal{L}_2 = \frac{1}{T-1}\sum_{t=1}^{T-1} \|\boldsymbol{z}_{t+1} - A(\boldsymbol{z}_t) - B(\boldsymbol{u}_t)\|_2^2, \quad \text{where} \quad \boldsymbol{u}_t = \mathrm{IDM}(\boldsymbol{z}_t, \boldsymbol{z}_{t+1}). \tag{2}$$

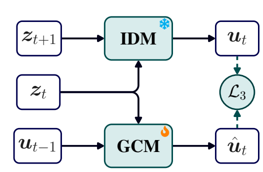

Figure 3: Action-matching training objective during phase 3. The GCM regresses onto the pseudo-action $\boldsymbol{u}_t$ generated by the frozen IDM.

This latent-space loss is highly efficient ($d_{\boldsymbol{z}} \ll H \times W \times C$) and prevents the model from wasting capacity on irrelevant pixel-level features, similar to [Assran et al., 2023]. That said, if phase 1 is skipped, pixel-space reconstruction terms can be used instead of $\mathcal{L}_2$. For instance, in our Moving MNIST experiments, jointly optimising phases 1 and 2 significantly accelerated convergence.

To regularise the dynamics models, we augment the training data distribution. First, we inject *temporally inverted* trajectories, denoted by the slice notation $\mathbf{o}_{T:1:-1}$, to encourage action symmetry. Second, we crop the videos and *pad the trajectory boundaries* to construct augmented $T-$length sequences $(\mathbf{o}_1, \mathbf{o}_1, \mathbf{o}_2, \ldots, \mathbf{o}_P, \mathbf{o}_P)$. By introducing these static transitions where $\boldsymbol{z}_{t+1} = \boldsymbol{z}_t$, we implicitly supervise the IDM to output a "do nothing" action, while constraining the FDM to act as the identity mapping under this null action.

Phase 3. In most downstream applications, future frames are unavailable, in which case we rely on a Generative Control Model (GCM), a causal sequence model, to auto-regressively predict actions $\boldsymbol{u}_t$ from $\boldsymbol{z}_t$ and $\boldsymbol{u}_{t-1}$ (see Figure 3 and Algorithm 1). This allows the model to autonomously generate plausible future frames, thereby transforming it into a frame-conditioned video generator or (behaviour) clone that can imitate an agent only from videos [Pomerleau, 1988].

The GCM is trained to match the IDM's action sequence via the objective $\mathcal{L}_3 = \frac{1}{T-1}\sum_{t=1}^{T-1}\|\boldsymbol{u}_t - \text{GCM}(\boldsymbol{z}_t, \boldsymbol{u}_{t-1})\|_2^2$.[4] We consider both recurrent architectures like LSTM [Hochreiter and Schmidhuber, 1997] and GRU [Cho et al., 2014] that only observe $\boldsymbol{z}_t$ and $\boldsymbol{u}_{t-1}$ at each time step, and Transformer-based architectures [Vaswani et al., 2017] that leverage all available $\boldsymbol{z}_{1:t}, \boldsymbol{u}_{1:t-1}$ pairs.

3.3 Implementation

All models are implemented in JAX using Equinox [Kidger and Garcia, 2021], optimised with Adam [Kingma and Ba, 2014] with a variable learning rate optimally tuned for each dataset and phase. NOVA is trained with a single RTX 4080 GPU with 16 GB of VRAM. Across all datasets, we use the same deep and narrow INR (6 MLP layers of width 12) preceded by Fourier coordinate encoding [Tancik et al., 2020], which we empirically found to be expressive enough to faithfully capture a wide variety of textures. The resulting latent dimension is compact, requiring as few as $d_{\boldsymbol{z}} = 961$ weights. To limit its information content, we constrain the action dimension to $d_{\boldsymbol{u}} = 4$ across all experiments unless stated otherwise. By default, we prioritise recurrent architectures for the GCM to limit attention to previous action-states, since these might leak information during intervention. These choices result in a total NOVA parameter count of approximately 40M. All results are reported on the test sets, predominantly using the trained GCM to generate actions for our interventions.[5] Further architectural details, along with hyperparameters, are presented in Section B.2.

4 Results and Discussion

Experiments are designed to demonstrate the performance of weight-space world models and video generators on readily available, yet surprisingly well-suited, unlabelled video datasets. Accordingly, we structure our results by dataset, dedicating a specific subsection to each. Additional results, including evaluations on further datasets, are provided in Section D.

4.1 Generation, Analysis, and Editing of Moving Digits

NOVA is a context-conditioned video generation model. *How much context does a video generation model need to capture true dynamics?* Figure 4 stress-tests this dependency by progressively restricting the GCM-based generation from 13 context frames down to 1. Remarkably, NOVA mirrors the ground truth (GT) dynamics near-perfectly; we only observe a meaningful divergence once the context window drops to 4 frames. Even under the extreme constraint of a single context frame, NOVA relies on its learned dynamics priors to synthesise coherent, physically plausible trajectories without mode collapse. We observe similarly crisp, artefact-free context-conditioned generation on both PhyWorld and WeatherBench (see Figures 14, 17 and 20 in Section D.2).

NOVA is resilient over long horizons. The true test of a world model, however, is its resilience over extended rollouts. In Section D.2, we push models trained on mere 20-step sequences to forecast 1,000 steps into the future. Under such a long horizon, the pixel-space LAPO baseline rapidly fails as it converges to a permanent dark figure, while the standard WM fails to commit to distinct digit identities with its mean-seeking average. In contrast, NOVA's formulation maintains structural integrity and object identity across the majority of its steps, suggesting that weight-space dynamics may be a necessary antidote to long-horizon mode collapse (see Figures 15 and 16 and Table 7).

NOVA induces a hierarchy of representations. We aim to provide empirical evidence that NOVA learns the structural hierarchy of background, foreground, and motion (denoted by $\bar{\boldsymbol{z}}, \boldsymbol{z}_t, \boldsymbol{u}_t$, respectively) without any supervision on the semantic meaning of these components. First, in all Moving MNIST sequences [Srivastava et al., 2015], the background is uniformly dark; our study finds that

[4] The required input $\boldsymbol{u}_0$ is computed from zero memory, with additional details in Section B.3.
[5] The performance of the GCM is indicative of that of its IDM teacher, which supplies pseudo-actions during phase 3.

Figure 4: **Robustness to minimal context.** Top row: ground truth (GT); bottom row: $L = 1$ context frame. Even under this extreme constraint, NOVA produces coherent, near-perfect predictions.

this background is absorbed into $\bar{z}$, as its direct rendering leads to black pixels (see Figure 11 in Section D). It therefore remains to be shown that $\boldsymbol{z}_t$ encodes the identity of dynamic objects, and that $\boldsymbol{u}_t$ dictates their motion rather than their appearance.

To show this, we design an experiment demonstrating the possibility of editing the identity without impacting the dynamics. Starting at $t = 11$, we compute a convex combination of $\boldsymbol{z}_t$ and $\boldsymbol{0}$ such that $\boldsymbol{z}_t \xrightarrow{t \to 20} \boldsymbol{0}$, while leaving GCM-extracted actions $\boldsymbol{u}_t$ free. As we observe in the top three rows on Figure 5, digit identities gradually morph towards a resting state (8s and 9s), all the while motion trajectories are preserved. We emphasise that this is achieved without any training supervision on digit identity. This finding is complemented by Figure 1, which showed that abruptly replacing $\boldsymbol{z}_t$ with an alien state does not perturb the original dynamics. These experiments establish that the A network acts as an object detector that processes foreground necessarily encoded in its only input $\boldsymbol{z}_t$.

Similarly, we interpolate $\boldsymbol{u}_t \xrightarrow{t \to 20} \boldsymbol{0}$ while leaving $\boldsymbol{z}_t$ free. As observed in the bottom three rows of Figure 5, digits decelerate and converge towards the canvas centre, while their visual identities are fully preserved throughout. This suggests that B is the component that processes instructions, necessarily encoded in its input $\boldsymbol{u}_t$. Together, these results underscore the capacity to intervene at arbitrary steps of the generation process.

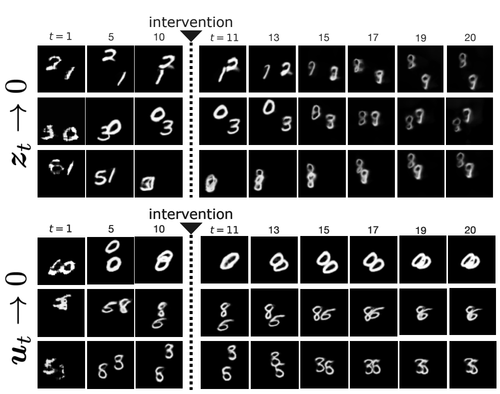

Figure 5: **Foreground and motion editing.** The top three rows, with $\boldsymbol{z}_t \to 0$, gradually morph digits in all sequences into 8s and 9s. The bottom three rows, with $\boldsymbol{u}_t \to 0$, force the digits to converge to the centre of the canvas.

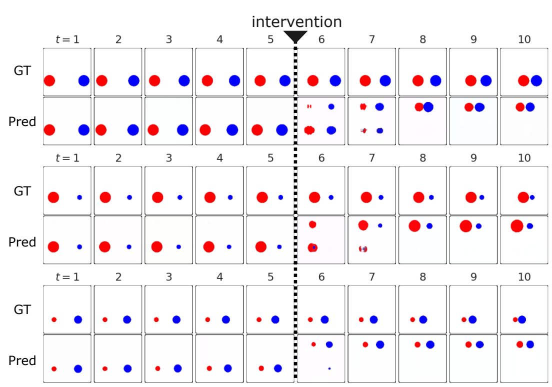

Figure 6: **Zero-shot motion retargeting via latent action substitution.** By replacing the native action on several PhyWorld sequences with an alien action after 5 steps, the model forces the original objects to inherit alien, "unphysical" trajectories.

Table 1: Comparison of additive and joint FDM performance on in-distribution (InD) and out-of-distribution (OOD) PhyWorld data. Samples are phase 3 videos generated from 3 context frames. Values are reported as mean $\pm$ standard deviation over the tested sequences.

Distribution	Model	Image Metrics			Physical Metrics		
		MSE ($\times 10^{-2}$) $\downarrow$	**PSNR** $\uparrow$	**SSIM** $\uparrow$	**Pos.** ($\times 10^{-2}$) $\downarrow$	**Mom.** $\downarrow$	**KE** $\downarrow$
InD	Additive	0.98 ± 0.71	23.12 ± 2.69	0.9404 ± 0.0219	2.61 ± 2.19	0.0548 ± 0.0408	0.0291 ± 0.1317
	Joint	1.08 ± 0.69	22.38 ± 2.04	0.9358 ± 0.0208	2.67 ± 2.53	0.0582 ± 0.0714	0.0456 ± 0.2112
OOD	Additive	3.20 ± 2.85	18.83 ± 4.81	0.9015 ± 0.0583	12.50 ± 8.99	0.4025 ± 0.5581	0.5351 ± 1.7191
	Joint	2.99 ± 2.30	17.83 ± 3.61	0.9015 ± 0.0547	9.67 ± 7.90	0.2093 ± 0.2243	0.0815 ± 0.2331

4.2 Zero-Shot Motion Retargeting and OOD Evaluation on PhyWorld

To rigorously evaluate the limits of NOVA's latent representation, we test it on the PhyWorld 30K collision validation set [Kang et al., 2025], which introduces ball radii and velocities far outside the training distribution. In this problem, the identity of a ball refers to its size, colour, and to some extent its y-position, which is constant throughout the video (see Section E). In Section D.2.2, we show that our model is **robust to latent state interventions** and can recalculate physically accurate post-collision velocities using the alien quantities. We ultimately show that latent actions are the central quantity for preserving physical consistency.

While the model maintains accurate collision dynamics across both in- and out-of-distribution settings, our primary interest lies in its capacity for action-controllable generation.

NOVA can abruptly retarget motion via action substitution. We test the orthogonality of our latent spaces by performing mid-sequence motion retargeting. We allow the system to evolve normally up to step $t = 5$, and then we abruptly substitute the native latent action $\boldsymbol{u}_t$ with an alien action extracted from a different sequence. While this is similar to moving digits depicted in Figure 5, we note that this abrupt intervention on PhyWorld, which we term as **zero-shot motion retargeting**, involves rigorous physics, raising important questions about the conservation of physical quantities such as momentum (Mom.) and kinetic energy (KE) [Kang et al., 2025].

Our results in Figure 6 indicate that our intervention forces the original balls to quickly adopt the motion trajectories of the alien sequence within less than two steps, deliberately violating the momentum conservation expected of their original masses. Crucially, the structural identity of the balls (radius and colour) remains perfectly preserved despite the unphysical motion. NOVA's ability to strictly obey the injected action without leaking structural features from the source sequence suggests orthogonality between weight-space identity and dynamic intent. As it allows users to seamlessly and privately[6] map arbitrary trajectories onto existing objects without corrupting visual identity, motion retargeting with unphysical dynamics is desirable, especially for applications involving digital content manipulation, stylistic or editorial control.

Ablation: additive vs. joint forward dynamics. In Table 1, we quantify the trade-offs of our decomposed architecture by comparing our additive FDM against a standard monolithic MLP that processes the concatenated vector $[\boldsymbol{z}_t, \boldsymbol{u}_t]$ jointly. While the joint architecture achieves lower physical errors on OOD configurations, it fundamentally destroys the content/dynamics decoupling required for the motion retargeting demonstrated above. Thus, our additive formulation accepts a meaningful but acceptable trade-off in exchange for disentanglement.

4.3 Zero-Shot Super-Resolution for Global Temperature Prediction

High-quality visuals are crucial for immersive virtual experiences and for detailed scientific analysis. In this section, we ask: *can we extract fine-grained meteorological insights from weather models simply by altering the rendering resolution?* To investigate this, we leverage NOVA's continuous spatial formulation. During training, the model evaluates the spatial field by sampling a small standard coordinate grid of $H \times W = 32 \times 64$ points. At inference,[7] however, we introduce a scaling factor s and resample the grid to a denser $(s \cdot H) \times (s \cdot W)$ resolution. When $s \gg 1$, this operation yields **zero-shot super-resolution** without requiring any specialised upsampling networks.

[6] This is private since it does not leak the identity of the objects that initiated the actions.

[7] To focus on upscaling quality, we disregard the GCM and rely on actions issued by the IDM to drive the video generation.

However, merely querying this denser grid risks aliasing if the network evaluates high-frequency features unsupported by the training resolution [Barron et al., 2021; Lindell et al., 2022]. To guarantee artefact-free upscaling, we implement a strict and simple **frequency-masking procedure** bounded by the Nyquist–Shannon theorem. By dynamically zeroing out any Fourier basis frequencies exceeding the Nyquist limit of the original training grid (e.g., $\xi^{y}_{\text{Nyquist}} = {}^{1}/{2\Delta y}$), we ensure the continuous field remains securely within its learned, unaliased manifold (detailed in Section D.3).

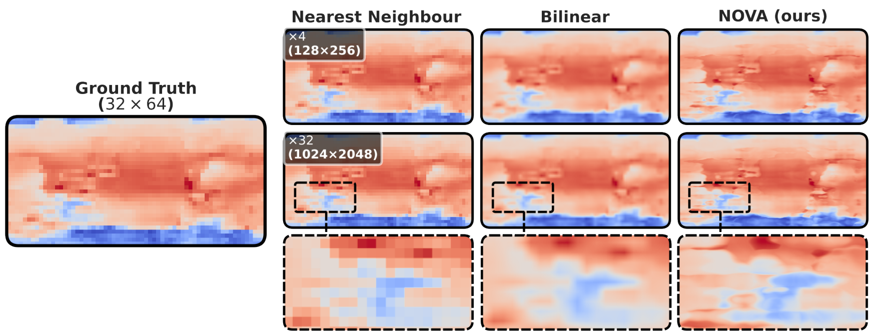

Figure 7: **Zero-shot super-resolution on WeatherBench.** Ground truth at native 32×64 resolution next to nearest-neighbour, bilinear interpolation, and NOVA at $\times 4$ and $\times 32$ scaling. NOVA avoids aliasing and preserves macro-structure without hallucinating high-frequency textures.

Figure 7 compares NOVA against nearest-neighbour and bilinear interpolation baselines on a single WeatherBench frame [Rasp et al., 2020] at three scaling factors. Our INR-based approach avoids the spurious discontinuities of nearest-neighbour and the artificial diffusion of bilinear interpolation at both downsampling and extreme upsampling scales.[8]

Quantitative results in Table 2 compare distributional fidelity and state preservation metrics across all strategies. We report mean and standard deviation metrics across all 24 time steps and across the batch dimension, focusing on the $\times 32$ upsampling resolution. While we observe that these distances are inherently narrow, the key insight is consistency: despite the tight margins, NOVA systematically matches or outperforms both nearest-neighbour (NN) and bilinear interpolation across every empirical distribution metric. Taken together with qualitative insights from Figure 7, it follows that, simply by re-rendering, NOVA can enhance the sharp, highly dynamic temperature distributions of evolving weather fronts without incurring artefacts. This capability has massive implications for localised extreme weather forecasting, where such precision impacts the lives of massive populations.

Table 2: Super-resolution ($\times 32$) evaluation across the test set. We employ a stratified sampling strategy over 69 batches to assess distributional and structural faithfulness to the original resolution. NOVA consistently achieves the lowest distributional drift.

Metric	**NN**	**Bilinear**	**NOVA**
$W_1 \downarrow$	0.137 ± 0.016	0.140 ± 0.015	$\mathbf{0.135 \pm 0.015}$
JSD $\downarrow$	0.229 ± 0.017	0.221 ± 0.022	$\mathbf{0.213 \pm 0.018}$
$B \downarrow$	0.056 ± 0.009	0.053 ± 0.012	$\mathbf{0.048 \pm 0.009}$
$d_{\mathcal{F}} \downarrow$	$\mathbf{0.427 \pm 0.065}$	$\mathbf{0.427 \pm 0.065}$	$\mathbf{0.427 \pm 0.067}$

5 Conclusion

We introduced NOVA, a world model and video generation framework that operates entirely in the weight space of an INR. By leveraging an analytical INR renderer, NOVA avoids the decoder bottleneck and achieves native resolution independence. Furthermore, it decouples background and foreground from motion substitution, enabling zero-shot latent state (and action) editing. Through experiments on diverse datasets, we demonstrate strong context-conditioned controllability and forecasting, operating at around 40M parameters on a single consumer GPU. Because the conceptual mechanism underlying NOVA's editing capabilities is applicable to standard latent action models, it could yield substantial benefits if integrated with mainstream world models. We foresee applications in various downstream tasks for digital artists and animators, for whom this uncompromising separation of motion and form could unlock a new paradigm of limitless expressive control.

[8] While NOVA natively supports upsampling to arbitrarily high resolutions, we cap our visualisation at $\times 32$, as further magnification yields diminishing perceptual returns. We show downsampled temperature maps in Figure 22.

Limitations and Future Work. While NOVA excels at controllable generation, its formulation leaves room for extensions: (i) continuous actions can include high-frequency artefacts (e.g., brightness), necessitating further regularisations for purer decoupling; (ii) OOD generalisation remains challenging, partly because NOVA does not feature the ability to leverage differential equations; and (iii) scaling to higher-dimensional INRs may demand parameter-efficient adaptations. We detail these limitations in Section E, highlighting them as exciting avenues for future work.

Broader Impact

Weight-space world models offer a more interpretable alternative to opaque abstract latent spaces by separating geometric semantics into background, content, and motion. While this transparency is highly beneficial for safety-critical applications, allowing researchers to audit model behaviour, identify biases, and apply safe interventions in fields like autonomous driving and robotics, it also introduces severe risks. The precise ability to intervene and edit video content could be easily exploited by bad actors to generate highly realistic deepfakes, manipulate visual evidence, and spread misinformation. To mitigate these dangers and promote community-driven oversight, we fully open-source the code, datasets, and model weights for both our proposed framework and the implemented baselines: `https://github.com/ddrous/nova`.

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Render, Don't Decode: Weight-Space World Models with Latent Structural Disentanglement

—*Supplementary Material*—

A Related Work

Our framework shares similarities with distinct fields and frameworks within the deep learning literature. We detail these major intersections below, while providing in Table 3 a high-level summary of how NOVA positively distinguishes itself.

World models and video generation. World models [Li et al., 2025b; Zhang et al., 2025b] have seen a massive surge in popularity in recent years. The classic paradigm of [Ha and Schmidhuber, 2018] uses a variational autoencoder to compress observations and an RNN to model latent transitions. DreamerV3 [Hafner et al., 2020] scales this with categorical representations and a joint-embedding world model. IRIS [Micheli et al., 2022] replaces the continuous latent with discrete tokens processed by a Transformer. Genie [Bruce et al., 2024] and V-JEPA [Bardes et al., 2023] learn from unlabelled video via latent-only objectives. These models all rely on a learned decoder. A complementary direction [He et al., 2025] fine-tunes video generation models to behave as world models. NOVA goes the other direction, as it starts with world modelling and recovers video generation capability through the GCM. LeWorldModel [Maes et al., 2026] demonstrates the benefits of end-to-end training without freezing encoders; NOVA achieves similar benefits by allowing for the possibility to combine its first two training phases, while additionally eliminating the decoder.

Latent action models (LAMs). Similar to reinforcement learning [Sutton et al., 1998], world models typically require explicit actions to inform their dynamics. When ground-truth actions are unavailable, latent action models (as described in Figure 8) infer these from consecutive observations. LAPO [Schmidt and Jiang, 2023] and LAPA [Ye et al., 2024] learn inverse dynamics models that extract latent actions bridging consecutive states. Moto [Chen et al., 2025] and AdaWorld [Gao et al., 2025] incorporate such models into robot learning pipelines. Garrido et al. [2026] demonstrate that continuous latent actions, although sparse and noisy, capture for in-the-wild actions learning, whereas discrete codebooks struggle. For deployment without future frames, RSSMs [Hafner et al., 2020], IRIS [Micheli et al., 2022], and CoWVLA [Yang et al., 2026] use autoregressive models to generate actions. Our GCM follows this principle using a causal sequence model.

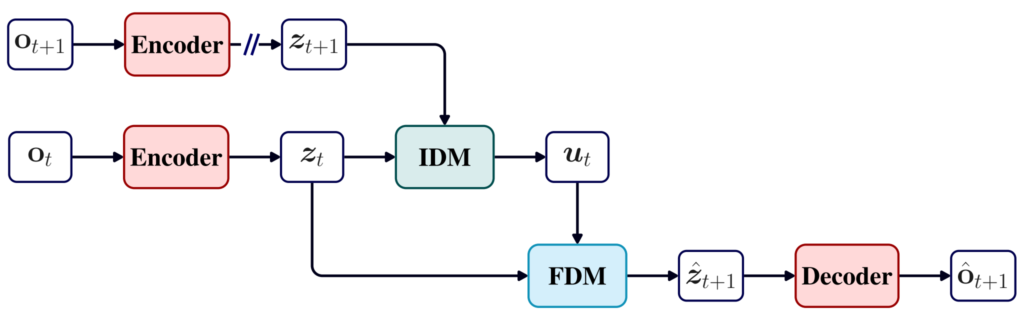


Figure 8: **Standard latent action model (LAM).** The IDM computes the action from consecutive latent states during training. This approach relies on a spatial decoder to project the latent state back into pixel space, creating a resolution bottleneck (e.g., [Schmidt and Jiang, 2023]). Note that the encoder and the decoder can be identity mappings as well, to match the setting of [Zhang et al., 2025a].

Disentanglement in video prediction. Decades of work aim to separate content from motion in video. DrNET [Denton et al., 2017] uses adversarial training to decompose frames into pose and content codes. DDPAE [Hsieh et al., 2018] employs structured probabilistic models to disentangle components with low-dimensional temporal dynamics. PhyDNet [Guen and Thome, 2020] introduces a PDE-constrained recurrent cell that explicitly separates physical dynamics from residual uncertainty. More recent work [Abdulsalam et al., 2026] disentangles static from first-order dynamic changes in a latent motion model. Critically, all these approaches impose disentanglement through auxiliary objectives or explicit structural priors applied on top of abstract latent spaces, and still require a learned decoder. In NOVA, disentanglement is a consequence of the weight-space geometry itself, requiring no additional loss terms.

Our work is particularly motivated by the fact that patch-based video representations [Lin et al., 2020], which power the majority of current video and world models, typically lack the human-style intuitiveness of object-centric decompositions [Daniel et al., 2026]. Because they struggle to capture such clean relationships between objects (which leaves them uninterpretable), they underperform, especially in out-of-distribution scenarios [Daniel et al., 2026]. Our work closes this gap by proposing weight-based representations, positioning itself as a crucial step for understanding complex scenes.

Implicit Neural Representations. INRs [Sitzmann et al., 2020] parameterise continuous signals via the weights of a coordinate-based MLP. Fourier features [Tancik et al., 2020] enable learning of high-frequency components. Dupont et al. [2022] introduce the term "functa", showing that meta-learned [Zintgraf et al., 2019; Nzoyem et al., 2025] INR weight vectors can serve as data representations for downstream tasks. In the 3D domain, 3D Gaussian Splatting [Kerbl et al., 2023] and work such as GWM [Lu et al., 2025] exploit *structured* spatial representations for world modelling. Our work extends this to 2D video dynamics, using the INR weights as the latent state that evolves under recurrent dynamics.

A fundamental advantage of INRs is their continuous formulation, encoding data as functional mappings rather than discrete spatial grids [Essakine et al., 2024]. This paradigm theoretically enables the representation of complex, high-resolution details with exceptional memory efficiency, circumventing the storage bottlenecks typical of high-dimensional video data. However, despite this compact capacity, standard INRs are notoriously susceptible to severe aliasing artefacts when evaluated at high resolutions [Barron et al., 2021]. Addressing this tension between memory efficiency and alias-free high-fidelity rendering is a central consideration in our work.

Weight-space sequence learning. [Li et al., 2025a; Nzoyem et al., 2026] demonstrate that the hidden state of a network can be parameterised as the weights of an auxiliary network, enabling (linear) weight-space sequence modelling. WeightCaster [Nzoyem, 2026] extends weight-space methods to out-of-support generalisation. NOVA is a world model that can be viewed as a non-linear generalisation of WARP [Nzoyem et al., 2026]. It mainly replaces the globally linear recurrence with a decomposed MLP dynamics model, and adds the base-weight stabilisation mechanism that makes training on complex visual signals tractable. Although designed for different applications, we compare both frameworks in further detail in Section D.5.

B Methodological Details

B.1 Motivation: INRs and The Decoder Bottleneck

Figure 9 contrasts abstract and structured representations. The abstract paradigm has three weaknesses as a world model latent space. First, the decoder typically consumes more parameters than the encoder [Bruce et al., 2024], making it a computational bottleneck. Second, the abstract latent space has no guaranteed semantic structure since disentangling background from foreground, or content from motion, requires explicit post-training alterations. Third, the renderer is resolution-locked: changing the output resolution requires retraining or expensive upsampling.

The structured paradigm eliminates these problems. The INR renderer is a fixed mathematical function, since querying any spatial coordinates requires no parameters. The weight vector z has a direct interpretation as it parameterises a continuous spatial function. And the continuous domain means we can query at any resolution by simply changing the coordinate grid.

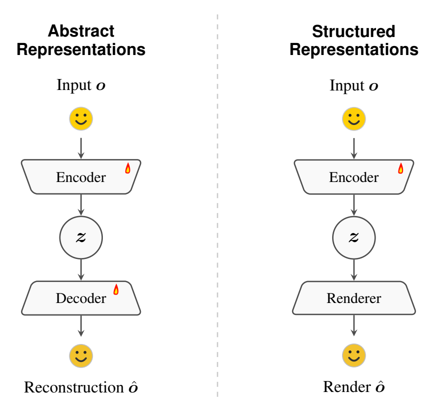


Figure 9: Two paradigms for signal encoding. **Abstract** encoders map input o to an opaque latent z requiring a trained decoder. **Structured** encoders map o to INR weights z, which are subsequently rendered analytically.

Table 3: **Capability comparison.** ✓= yes; ✗= no; ≈= partial or config-dependent. † = requires known action labels at training time. Entries reflect standard published configurations. NOVA is uniquely positioned among representative baselines by simultaneously achieving decoder-free and resolution-independent rendering, latent content-motion disentanglement, zero-shot super-resolution, and action-label-free training directly from video data, all while remaining efficient enough to be trained on a single consumer GPU.

	NOVA (ours) [Figure 2]	Standard WM [Figure 8]	WARP [Nzoyem et al., 2026]	DreamerV3† [Hafner et al., 2020]	GENIE [Bruce et al., 2024]	V-JEPA [Bardes et al., 2023]	IRIS† [Micheli et al., 2022]	PredRNN [Wang et al., 2022]	PhyDNet [Guen and Thome, 2020]	LeWorldModel† [Maes et al., 2026]
Architecture										
Decoder-free rendering	✓	✗	✓	✗	✗	✓	✗	✗	✗	✓
Resolution-independent rendering	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗
Single-GPU training	✓	✓	✓	✗	✗	✗	✓	✓	✓	✗
Disentanglement & Control										
Latent content–motion disentanglement	✓	≈	✗	✗	✗	✗	✗	✗	✗	✗
Structural A/B dynamics decomposition	✓	≈	✗	✗	✗	✗	✗	✗	≈	✗
Zero-shot content/motion retargeting	✓	≈	≈	✗	✗	✗	✗	✗	✗	✗
Latent action model (actions from video)	✓	✓	≈	✗	✓	✗	✗	✗	✗	✗
Continuous action space	✓	✓	✓	✓	✗	✗	✗	✗	✗	✓
Generation										
Zero-shot super-resolution	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗
Context-conditioned forecasting	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Autonomous generation (no test-time actions)	✓	✓	≈	✓	✓	✓	✓	✓	✓	✓
Artefact-free rendering	✓	≈	✓	≈	≈	✓	≈	≈	≈	✓
Domain coverage										
Continuous geospatial physics	✓	✓	✓	✗	✗	≈	✗	✓	✓	✗
Goal-directed discrete control	✓	✓	≈	✓	✓	✗	✓	✗	✗	✓

By leveraging Fourier features [Tancik et al., 2020], these representations provide a differentiable, 3D-consistent bridge between latent space and pixels. Fourier mapping allows the network to overcome spectral bias, capturing the high-frequency details and sharp gradients (e.g., fine textures, localised weather fronts) that standard MLPs often blur. This continuous, analytical framework enabled zero-shot super-resolution. Such scalability is critical for applications like global weather forecasting, where a single pixel must represent vastly different scales without loss of structural integrity.

Finally, structured representations can be significantly more compact than raw pixel arrays [Mostajeran et al., 2025]. By encoding complex data into 1D weight vectors, they can be processed with generic MLPs rather than specialised architectures like CNNs or RNNs, thus simplifying the pipeline.

B.2 Architecture & Hyperparameters

Figure 10 provides a detailed view of key NOVA components at the configuration used in WeatherBench experiments (see Table 5):

(a) **Implicit Neural Representation.** A 6-layer[9] MLP of width 12 preceded by a Fourier coordinate encoding with 6 frequency bands. The weights and biases of all 5 layers are combined as $\bar{\boldsymbol{z}} + \boldsymbol{z}_t$ at render time. Non-batched inputs are $(\mathrm{x}, \mathrm{y}) \in [-1, 1]^2$; the output is the pixel value at the queried coordinate.

(b) **CNN Encoder.** A 5-layer strided convolutional backbone (stride 2, channels 64/128/256/512, ReLU activations) followed by a flatten and linear projection to $\mathbb{R}^{d_{\boldsymbol{z}}}$. We use 3×3 kernels.

(c) **Inverse Dynamics Model.** A 4-layer MLP of width $d_{\boldsymbol{z}}$ taking concatenated $(\boldsymbol{z}_t, \boldsymbol{z}_{t+1})$ as input and outputting $\boldsymbol{u}_t \in \mathbb{R}^{d_{\boldsymbol{u}}}$.

(d) **Generative Control Model.** A causal Transformer with 4 blocks (8 attention heads, MLP ratio 4) that processes a sequence of concatenated $\{\boldsymbol{z}_\tau, \boldsymbol{u}_\tau\}_{1 \leq \tau \leq t}$ pairs, with learned positional embeddings. We set the values corresponding to $\boldsymbol{u}_t$ as 0 in the input sequence.

[9] Note that the output layer is included in the layer count; meaning that 5 layers corresponds to a depth of 4, following Equinox's convention [Kidger and Garcia, 2021].

(e) **Forward Dynamics Model.** Two independent 4-layer MLPs: A of width $2d_{\boldsymbol{z}}$ taking $\boldsymbol{z}_t$ as input, and B of width $2d_{\boldsymbol{z}}$ taking $\boldsymbol{u}_t$ as input. Their outputs are summed to produce $\boldsymbol{z}_{t+1}$.

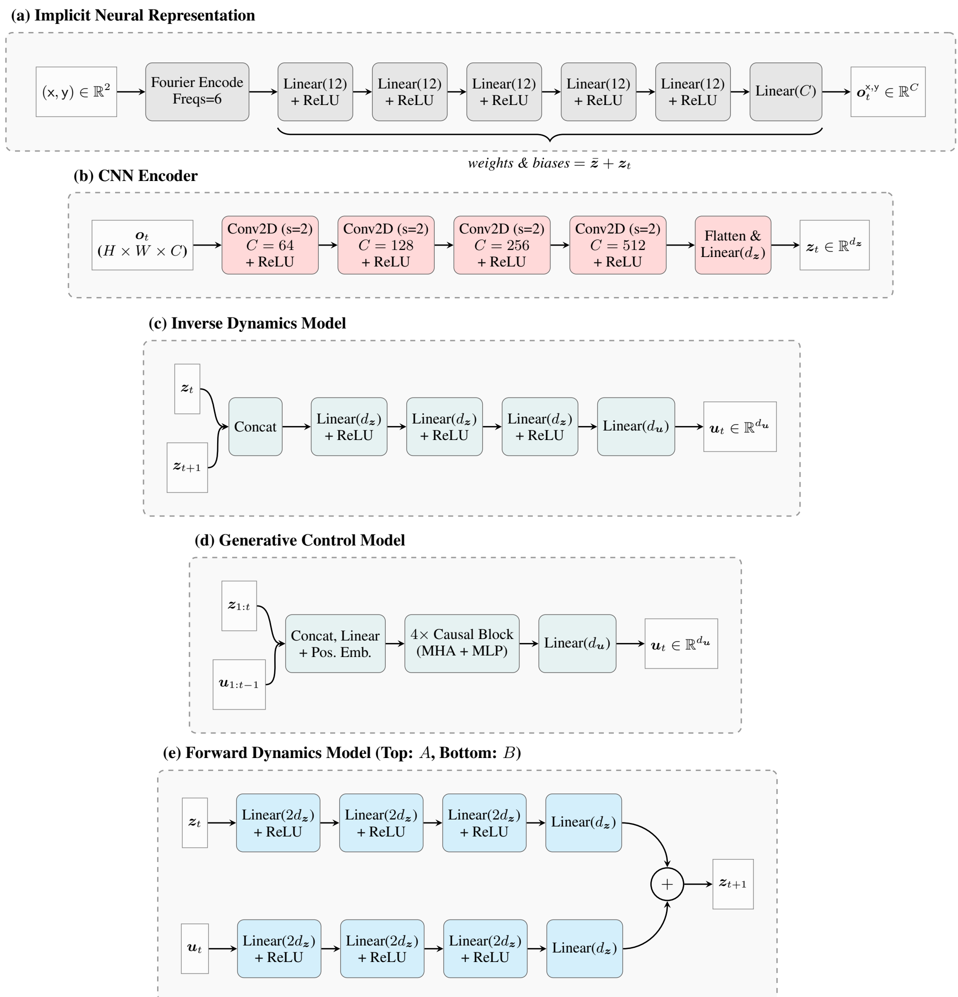


Figure 10: **Detailed NOVA components** as used on the WeatherBench dataset, corresponding to the experimental configuration ($d_{\boldsymbol{z}} = 961$, $d_{\boldsymbol{u}} = 16$, and $C = 1$).

Architecture of the GCM. We employ Transformer architectures exclusively for the WeatherBench and MiniGrid datasets, whereas we utilise a GRU for Moving MNIST and an LSTM for PhyWorld. Because the latter two datasets serve as the basis for our intervention experiments, we prioritise attention-free recurrent architectures to prevent the model from inadvertently attending to historical, uncorrupted states. At each time step, the recurrent core (LSTM or GRU) receives the concatenated latent observation and previous action as input to update its internal hidden memory state. This updated memory state is then passed to an MLP decoder, which projects the concatenated hidden state and current latent observation into the target action space. Successfully accommodating this breadth of sequence-modelling backbones underscores the versatility of our framework, demonstrating that it is robust and agnostic to the choice of GCM architecture.

Structural inductive biases. It is important to note that while NOVA successfully disentangles background, object identity, and motion without any semantic supervision (i.e., no explicit labels nor ground-truth masks are provided), this emergence is facilitated by minimal structural bottlenecks rather than rigid inductive biases. Specifically, isolating $\bar{\boldsymbol{z}}$ from the temporal dynamics naturally encourages the absorption of static elements, while the additive structure of the FDM ($A(\boldsymbol{z}_t)+B(\boldsymbol{u}_t)$) provides a simple factorisation of state representations and temporal transitions. Furthermore, employing an Implicit Neural Representation supplies a basic coordinate-based spatial grounding without enforcing domain-specific priors.

B.3 Context-Conditioned Video Generation

To generate videos with up to T_{inf} frames, a context ratio $\rho \in [0, 1]$ controls the fraction of steps in which the reference ground-truth next frame is available. For $t/T \leq \rho$, the IDM provides the action; for $t/T > \rho$, the GCM takes over. Setting $\rho = 1$ gives oracle world modelling; $\rho = 0$ gives single-frame-conditioned video generation. Algorithm 1 details the full procedure.

Algorithm 1 NOVA Context-Conditioned Video Generation

Require: Ref. video $\mathbf{o}_{1:T}$, inference steps T_{inf}, context ratio $\rho \in [0, 1]$, coord. grid (X, Y)

$\boldsymbol{z}_1 \leftarrow \text{Enc}(\mathbf{o}_1)$
$\boldsymbol{m}_1 \leftarrow \text{GCM}_{\text{init}}()$ ▷ Initialise memory buffer/state
for $t = 1$ **to** T_{inf} **do**
  $\boldsymbol{\theta}_t \leftarrow \texttt{unflat}(\bar{z} + \boldsymbol{z}_t)$
  $\hat{\mathbf{o}}_t \leftarrow \text{MLP}_{\boldsymbol{\theta}_t}(\mathsf{X}, \mathsf{Y})$ ▷ Render current frame
  if $t/T < \rho$ **then**
    $\boldsymbol{u}_t \leftarrow \text{IDM}(\boldsymbol{z}_t,\ \text{Enc}(\mathbf{o}_{t+1}))$ ▷ Extract action from context
  else
    $\boldsymbol{u}_t \leftarrow \text{GCM}_{\text{decode}}(\boldsymbol{m}_t,\ \boldsymbol{z}_t)$ ▷ Autoregressively predict action
  end if
  $\boldsymbol{m}_{t+1} \leftarrow \text{GCM}_{\text{encode}}(\boldsymbol{m}_t,\ \boldsymbol{z}_t,\ \boldsymbol{u}_t)$ ▷ Update memory with chosen action
  $\boldsymbol{z}_{t+1} \leftarrow A(\boldsymbol{z}_t) + B(\boldsymbol{u}_t)$ ▷ Forward dynamics step
end for
return $\hat{\mathbf{o}}_{1:T_{\text{inf}}}$

Managing memory during generation. The internal memory state $\boldsymbol{m}_t$ presented in Algorithm 1 takes different forms depending on the architecture: it is a fixed-size hidden vector in the recurrent architectures, but a sequence buffer (matrix) in the Transformer. We define three fundamental operations to abstract away internal GCM processing: *init*, *decode*, and *encode*.

For the **recurrent architectures** (GRU, LSTM), memory is initialised as a zero vector, $\boldsymbol{m}_1 = \mathbf{0} \in \mathbb{R}^{d_{\text{rnn}}}$. During *encode*, the memory is recursively updated through the recurrent cell:

$$\boldsymbol{m}_{t+1} = \text{RecurentCell}([\boldsymbol{z}_t, \boldsymbol{u}_t], \boldsymbol{m}_t). \tag{3}$$

The *decode* operation concatenates memory and latent state to predict the action via a MLP:

$$\boldsymbol{u}_t = \text{MLP}_{\text{decode}}([\boldsymbol{m}_t, \boldsymbol{z}_t]) \tag{4}$$

For the **Transformer architecture**, memory is initialised as the zero matrix $\boldsymbol{m}_1 = \mathbf{0} \in \mathbb{R}^{T \times d_{\text{transformer}}}$. During *encode*, the context token is computed using the current latent state and the newly predicted action. The memory is updated by permanently assigning this token to the t-th row of the buffer:

$$\boldsymbol{m}_{t+1}[i, :] = \begin{cases} \boldsymbol{W}_{\text{in}}[\boldsymbol{z}_t, \boldsymbol{u}_t] + \boldsymbol{b}_{\text{in}} & \text{if } i = t \\ \boldsymbol{m}_t[i, :] & \text{otherwise.} \end{cases} \tag{5}$$

Conversely, during *decode*, because $\boldsymbol{u}_t$ is not yet available, a query token is generated using a zero-padded action vector, $\boldsymbol{q}_t = \boldsymbol{W}_{\text{in}}[\boldsymbol{z}_t, \mathbf{0}] + \boldsymbol{b}_{\text{in}}$, and inserted into the t-th row of $\boldsymbol{m}_t$ to form a query buffer $\tilde{\boldsymbol{m}}_t$. This buffer is added to learned positional embeddings $\boldsymbol{E}_{\text{pos}}$ and processed through causally-masked Multi-Head Attention (MHA) and MLP blocks. The action is then linearly projected from the t-th output vector:

$$\boldsymbol{u}_t = \boldsymbol{W}_{\text{out}}\, \text{Transformer}(\tilde{\boldsymbol{m}}_t + \boldsymbol{E}_{\text{pos}})_t + \boldsymbol{b}_{\text{out}}. \tag{6}$$

In both architectures, the generation of the action $\boldsymbol{u}_t$ is causal. At any given timestep t, the GCM always sees the sequence of states up to the current step, $\boldsymbol{z}_{1:t}$, and the sequence of actions from the past, $\boldsymbol{u}_{1:t-1}$. This causal isolation justifies our simplified notations $\text{GCM}(\boldsymbol{z}_t, \boldsymbol{u}_{t-1})$ and $\text{GCM}(\boldsymbol{z}_{1:t}, \boldsymbol{u}_{1:t-1})$ utilised from Figure 3 onwards.

B.4 Zero-Shot Content and Motion Retargeting

Because the context-conditioned video generation process is a causal system, we can perform zero-shot content and motion retargeting, either independently or simultaneously, through targeted inference-time interventions (see Algorithm 2). To perform **content retargeting** at specific time steps included in the intervention set $\mathcal{I}$,[10] we intervene just before the forward dynamics step by substituting the current state $\boldsymbol{z}_t$ with an external alien state $\boldsymbol{z}_t^{\text{alien}}$.[11] Conversely, for **motion retargeting**, we intervene on the control variables by swapping the generated action $\boldsymbol{u}_t$ for an alien action $\boldsymbol{u}_t^{\text{alien}}$. Both interventions can be toggled as needed, operating cleanly within the existing generation pipeline without requiring any additional training or overhead.

Algorithm 2 Content and/or Motion Retargeting

Require: Reference video $\mathbf{o}_{1:T}$, inference steps T_{inf}, context ratio $\rho \in [0, 1]$, coordinate grid (X, Y), alien states $\boldsymbol{z}^{\text{alien}}$, alien actions $\boldsymbol{u}^{\text{alien}}$, intervention set $\mathcal{I}$

$\boldsymbol{z}_1 \leftarrow \text{Enc}(\mathbf{o}_1)$
$\boldsymbol{m}_1 \leftarrow \text{GCM}_{\text{init}}()$
for $t = 1$ **to** T_{inf} **do**
    $\boldsymbol{\theta}_t \leftarrow \texttt{unflat}(\bar{z} + \boldsymbol{z}_t)$
    $\hat{\mathbf{o}}_t \leftarrow \text{MLP}_{\boldsymbol{\theta}_t}(\mathsf{X}, \mathsf{Y})$
    if $t/T < \rho$ **then**
        $\boldsymbol{u}_t \leftarrow \text{IDM}(\boldsymbol{z}_t,\, \text{Enc}(\mathbf{o}_{t+1}))$
    else
        $\boldsymbol{u}_t \leftarrow \text{GCM}_{\text{decode}}(\boldsymbol{m}_t,\, \boldsymbol{z}_t)$
    end if
    $\boldsymbol{m}_{t+1} \leftarrow \text{GCM}_{\text{encode}}(\boldsymbol{m}_t,\, \boldsymbol{z}_t,\, \boldsymbol{u}_t)$
    if $t \in \mathcal{I}$ **then**
        $\boldsymbol{z}_t \leftarrow \boldsymbol{z}_t^{\text{alien}}$ ▷ Content intervention!
        $\boldsymbol{u}_t \leftarrow \boldsymbol{u}_t^{\text{alien}}$ ▷ Motion intervention!
    end if
    $\boldsymbol{z}_{t+1} \leftarrow A(\boldsymbol{z}_t) + B(\boldsymbol{u}_t)$
end for
return $\hat{\mathbf{o}}_{1:T_{\text{inf}}}$

C Implementation and Experimental Details

C.1 Datasets Properties

We evaluate our framework across diverse datasets, testing predictive capability, visual fidelity, and dynamic disentanglement across varied domains. Dataset properties are summarised in Table 4.

(1) **Moving MNIST** [Srivastava et al., 2015] assesses deterministic spatial-temporal forecasting and boundary sharpness over time; sequences ($T = 20$) show two digits bouncing on a 64×64 grayscale canvas. We use 8,000 training and 2,000 testing sequences.

(2) **PhyWorld 30K** [Kang et al., 2024] evaluates modelling of complex, non-linear multi-body physical interactions. Rendered at 128×128 RGB resolution, we use approximately 26,066 training sequences and 1,635 testing sequences containing both in-distribution and out-of-distribution characteristics (ball radii and velocities).

(3) **WeatherBench** [Rasp et al., 2020] tests continuous, global-scale real-world dynamics, popular as a benchmark for data-driven medium-range forecasting. The target temperature variable is standard-scaled using training statistics. We inherit the chronological split common in the literature: training (1979-2015), validation (2016), and testing (2017-2018).

(4) **MiniGrid** [Chevalier-Boisvert et al., 2023] challenges the model to extract discrete control policies from visual data without explicit action labels. We generated offline goal-directed navigation sequences using a Breadth-First Search (BFS) optimal policy, rendered as $72 \times 72 \times 3$.

[10] One single intervention step (i.e., $|\mathcal{I}| = 1$) can be enough for successful motion/content retargeting applicable to all subsequent steps, as we achieve on Moving MNIST.

[11] Note that all alien frame encodings $\boldsymbol{z}_t^{\text{alien}}$ could be identical, i.e., $\boldsymbol{z}_t^{\text{alien}} = \boldsymbol{z}_{t+1}^{\text{alien}}, \forall\, t = 1, \ldots, T_{\text{inf}} - 1$.

Table 4: Dataset properties and evaluation splits. We abbreviate phase 1, phase 2, and phase 3 as p1, p2, and p3, respectively. NA is used when phase 1 was skipped, meaning world model-adjacent components (Enc, IDM, and FDM) were trained end-to-end. The star (*) indicated that a test set was not available; in which case the prescribed PhyWorld validation set was used for testing.

Dataset	T	$H \times W \times C$	**Train size**	**Test size**	**Batch sizes (p1, p2, p3)**
Moving MNIST	20	$64 \times 64 \times 1$	8000	2000	NA, 256, 256
PhyWorld 30K	32	$128 \times 128 \times 3$	26066	1635*	16, 64, 64
WeatherBench	24	$32 \times 64 \times 1$	324313	17497	8, 128, 256
MiniGrid	10	$72 \times 72 \times 3$	8000	2000	16, 24, 24

C.2 NOVA & Baseline Implementations

Hyperparameters were manually tuned based on empirical performance on the validation sets when available. Most batch sizes specified in Table 4, along with a deep and narrow INR layers (which reduced the number of parameters $d_z = 961$ or 987) meant the memory consumption was kept low across all datasets, never exceeding 4GB. This small memory usage was also made possible by the fact that we used the stop-gradient operator as described in Figure 2, while instructing JAX to recompute all IDM- and FDM-related quantities during its backward pass. We use a learning rate of 10^{-4} for MNIST, and 10^{-5} for the majority of other datasets and phases. Unless otherwise specified, we generate videos and do interventions using 10 context frames for Moving MNIST and 3 for PhyWorld, as suggested in their respective papers. In the discrete NOVA implementation described in Section D.4 for MiniGrid, we used a codebook of 200 vectors, each of dimension 2; the same codebook is reused by the GCM when performing behaviour cloning. Further NOVA training details are provided in Table 5.

We implement the Standard WM in similar conditions to NOVA, even matching its batch sizes. As for the encoder-decoder-free LAPO [Schmidt and Jiang, 2023], we adapt the code from its official repository, readily available and downloaded from `https://github.com/schmidtdominik/LAPO`.

Table 5: Parameter count across datasets. "Vector" describes the dimension of one-dimension latent vectors, while "Submodel" describes submodel components within NOVA. "Total Params" counts all learnable submodel parameters, plus the length of the isolated $\bar{z}$. To prevent confusion, we present configurations for the MiniGrid implementation with continuous actions. Across all datasets, total learnable parameters amount to roughly 40 million.

Dataset	**Vector**		**Submodel**				**Total Params**
	$d_{\boldsymbol{u}}$	$d_{\boldsymbol{z}}$	$d_{\text{Enc.}}$	d_{FDM}	d_{IDM}	d_{GCM}	
Moving MNIST	4	961	9423297	20338604	2776333	1251588	33790783
PhyWorld 30K	4	987	16561398	21453432	3903589	1597444	43516850
WeatherBench	16	961	14070658	20361668	3712359	3421712	41567358
MiniGrid	4	987	6707190	21453432	3903589	1597444	33662642

Table 6: NOVA wallclock training times across datasets. NA indicates that phases 1 and 2 were performed jointly.

Dataset	**Train. Times (hours)**		
	Phase 1	**Phase 2**	**Phase 3**
Moving MNIST	NA	7.0	1.2
PhyWorld 30K	12.8	8.1	5.5
WeatherBench	0.9	2.4	1.0
MiniGrid	3.3	16.4	1.4

C.3 Evaluation Metrics

Conventional metrics. We use standard pixel-level metrics such as Mean Squared Error (MSE) for direct comparison during training, validation, and testing. Mean Absolute Error (MAE) is used during the latter two stages.

To complement this on the **PhyWorld** physics datasets, we extract the centre-of-mass position $x_t^{(i)}$ of each ball $i \in \{1, 2\}$ from each predicted frame and estimate the velocity via finite differences $v_t^{(i)} = (x_t^{(i)} - x_{t-1}^{(i)})/\Delta t$, with $\Delta t = 0.1$. Ball masses are set to $m_i = r_i^2$, proportional to the squared radius provided by the dataset's initial conditions. We then evaluate total linear momentum $\mathrm{Mom}_t = m_1 v_t^{(1)} + m_2 v_t^{(2)}$ and total kinetic energy $\mathrm{KE}_t = \frac{1}{2} m_1 (v_t^{(1)})^2 + \frac{1}{2} m_2 (v_t^{(2)})^2$, averaging across time and reporting the mean absolute error against the same quantities computed from ground truth positions issued by Kang et al. [2024].

Ball positions are extracted from predicted frames via colour thresholding. Each frame $\hat{\boldsymbol{o}}_t \in [0, 1]^{H \times W \times 3}$ is segmented into two binary masks: a red-ball mask, selecting pixels where the R channel exceeds $\tau = 0.8$ while G and B remain below τ, and a blue-ball mask, selecting pixels where B exceeds τ while R and G remain below τ. The centre-of-mass position of each ball is then computed as the mean pixel coordinate over the corresponding mask and normalised to $[0, 1]$ by dividing by the frame dimensions W and H respectively. If a mask is empty at time t, indicating the ball has left the field of view or is occluded, the position is set to the last known position at $t - 1$.

Distributional and structural metrics. Beyond trajectory-level agreement, we further assess structural and distributional consistency at the frame level through the metrics below.

Let $\hat{\boldsymbol{o}}_t \in [0, 1]^{H \times W \times C}$ denote the predicted frame at time step t, and let $\hat{p}_t$ be its empirical pixel-intensity distribution, estimated via a histogram over $[0, 1]$. We aim to quantify *identity drift*, stemming from the fact that a faithful world model should preserve the visual identity of each object even as it undergoes rigid motion; pixel-distribution metrics are invariant to such motion by construction. To quantify structural and distributional consistency over time, all metrics compare $\hat{\boldsymbol{o}}_t$ (or its histogram $\hat{p}_t$) against a ground truth $\boldsymbol{o}_t$ (or $\hat{p}$).

We remark that for long horizon forecasting tasks performed on Moving MNIST where ground truth beyond $T = 20$ is not available, we use, in place of p_t, the aggregate empirical distribution p_{emp} pooled over all available GT frames, i.e., $p_t = p_{\mathrm{emp}}, \forall\, t > 1$. For pixel-based metrics such as SSIM and PSNR, we solely rely on the first frame $\boldsymbol{o}_1$ because spatially averaging GT frames would introduce motion-blur artefacts that corrupt the structural reference.

All histogram-based metrics use N=64 equal-width bins; Wasserstein-1, JSD distance, and Bhattacharyya are computed via `scipy.stats.wasserstein_distance` and `scipy.spatial.distance.jensenshannon`, with bin centres as support points for W_1; SSIM and PSNR use `skimage.metrics` against $\boldsymbol{o}_t$; the FFT metric is computed via `numpy.fft.fft2` with the reference spectrum averaged over all GT frames for motion robustness.

- ***Wasserstein-1 distance*** (W_1 , $\downarrow$).

$$W_1(p_t, \hat{p}_t) = \inf_{\gamma \in \Gamma(p_t, \hat{p}_t)} \mathbb{E}_{(u,v)\sim\gamma}\big[|u - v|\big] = \int_0^1 \big|F_t(x) - \hat{F}_t(x)\big| \,\mathrm{d}x,$$

  where F_t and $\hat{F}_t$ are the CDFs of p_t and $\hat{p}_t$, respectively. The closed-form CDF reduction holds in one dimension [Vallender, 1974].

- ***Jensen–Shannon Divergence*** (JSD , $\downarrow$).

$$\mathrm{JSD}(p_t, \hat{p}_t) = \sqrt{\tfrac{1}{2} D_{\mathrm{KL}}(p_t \| m) + \tfrac{1}{2} D_{\mathrm{KL}}(\hat{p}_t \| m)},$$

  where $m = \frac{1}{2}(p_t + \hat{p}_t)$ and $D_{\mathrm{KL}}(p\|q) = \sum_b p_b \ln(p_b/q_b)$.

- ***Bhattacharyya distance*** (B , $\downarrow$).

$$B(p_t, \hat{p}_t) = -\ln \sum_{b=1}^{N} \sqrt{p_{t,b}\, \hat{p}_{t,b}},$$

  where the sum inside the logarithm is the Bhattacharyya coefficient, and N the number of bins; $B = 0$ iff $p_t = \hat{p}_t$, and $B \to \infty$ as the distributions become disjoint.

- ***Structural Similarity Index*** (SSIM, ↑).

$$\mathrm{SSIM}(\boldsymbol{o}_t, \hat{\boldsymbol{o}}_t) = \frac{\left(2\mu_o\, \mu_{\hat{o}} + C_1\right)\left(2\sigma_{o\hat{o}} + C_2\right)}{\left(\mu_o^2 + \mu_{\hat{o}}^2 + C_1\right)\left(\sigma_o^2 + \sigma_{\hat{o}}^2 + C_2\right)},$$

  where all statistics are computed within a sliding 11×11 Gaussian window ($\sigma_w = 1.5$): μ_o and $\mu_{\hat{o}}$ denote the local means, σ_o^2 and $\sigma_{\hat{o}}^2$ the local variances, and $\sigma_{o\hat{o}}$ the local covariance of $\boldsymbol{o}_t$ and $\hat{\boldsymbol{o}}_t$; the reported score is the mean of this map over all window positions. The stabilisation constants are $C_1 = (K_1 L)^2$ and $C_2 = (K_2 L)^2$, with $K_1 = 0.01$, $K_2 = 0.03$, and L the pixel dynamic range (e.g. $L = 1$ for images normalised to $[0, 1]$). $\mathrm{SSIM} \in [-1, 1]$, with 1 indicating perfect structural agreement.

- ***Peak Signal-to-Noise Ratio*** (PSNR, ↑).

$$\mathrm{PSNR}(\boldsymbol{o}_t, \hat{\boldsymbol{o}}_t) = 10 \log_{10}\left(\frac{1}{\mathrm{MSE}(\boldsymbol{o}_1, \hat{\boldsymbol{o}}_t)}\right) \text{ [dB]}, \qquad \mathrm{MSE}(\boldsymbol{a}, \boldsymbol{b}) = \frac{1}{HWC}\|\boldsymbol{a} - \boldsymbol{b}\|_F^2,$$

  where the peak value $\mathrm{MAX} = 1$ for the normalised range $[0, 1]$ has been substituted. Like SSIM, PSNR is not motion-invariant and is included as a complementary pixel-fidelity reference.

- ***FFT magnitude distance*** ($d_{\mathcal{F}}$, ↓).

$$d_{\mathcal{F}}(\mathsf{T}, \hat{\boldsymbol{o}}_t) = \frac{1}{HW}\left\|\frac{1}{|\mathsf{T}|}\sum_{t' \in \mathsf{T}} \left|\mathcal{F}(\boldsymbol{o}_{t'})\right| - \left|\mathcal{F}(\hat{\boldsymbol{o}}_t)\right|\right\|_F,$$

  where $\mathcal{F} : \mathbb{R}^{H\times W} \to \mathbb{C}^{H\times W}$ denotes the 2-D DFT, $|\cdot|$ the element-wise complex modulus, T the set of all GT frame indices, and $\|\cdot\|_F/(HW)$ the size-normalised Frobenius norm. While global rigid translation perfectly preserves the magnitude spectrum, scenes with multiple independent entities may exhibit phase interference. Averaging over T mitigates these oscillating cross-terms, yielding a robust spectral reference of the underlying object structures. This metric is only computed for $C = 1$.

D Additional Results

D.1 Latent Structural Disentanglement

Background absorption by $\bar{\boldsymbol{z}}$. In all Moving MNIST sequences, the background is uniformly dark. As seen in Figure 11, we observe that $\bar{\boldsymbol{z}}$ naturally absorbs this, indicating that the per-frame offsets $\boldsymbol{z}_t$ have near-zero contribution to background pixels.

The utility of $\bar{\boldsymbol{z}}$ extends beyond structural disentanglement. In particular, sequences with dynamic backgrounds or moving cameras, such as in-the-wild videos [Garrido et al., 2026], could mean that a single dataset-wide $\bar{\boldsymbol{z}}$ is not able to perfectly isolate all non-dynamic components. In such settings, we hypothesise that $\bar{\boldsymbol{z}}$ still acts as an important weight-space anchor, as demonstrated by our ablation in Figure 28. By framing the temporal predictions as small residuals $\boldsymbol{z}_t$ relative to $\bar{\boldsymbol{z}}$, analogous to residual learning [He et al., 2016], we stabilise optimisation and prevent the network from collapsing into local minima. Consequently, even in complex scenes where background disentanglement is challenging, the residual anchoring provided by $\bar{\boldsymbol{z}}$ may remain beneficial for efficient learning.

The A component processes content identity. Figure 12 illustrates a controlled intervention on Moving MNIST. After a context phase ($t \leq 10$), we replace $\boldsymbol{z}_t$ with a convex combination towards $\mathbf{0}$ (the zero offset vector), while keeping $\boldsymbol{u}_t$ free from the IDM. The zero vector corresponds to the base network $\bar{\boldsymbol{z}}$ alone, which renders a "prototype" or dataset-average scene. Feeding this interpolated state through A gradually morphs the digit identity towards lightly-shaded 8s and 9s, while preserving the motion trajectory computed by the GCM. Since perturbing $\boldsymbol{z}_t$ changes *what* appears, not *where* it moves, this demonstrates that $A(\boldsymbol{z}_t)$ processes the semantic identity of the objects.

The B component processes spatial motion. Figure 13 shows the complementary intervention: keeping $\boldsymbol{z}_t$ free while interpolating $\boldsymbol{u}_t$ towards $\mathbf{0}$. The digits progressively decelerate and converge towards the centre of the canvas. Their visual identity is fully preserved throughout. This confirms that $B(\boldsymbol{u}_t)$ processes motion instructions but carries no information about object identity.

Figure 11: Background isolation on Moving MNIST. Left: one frame from the dataset, rendered with $\bar{z} + z_t$; Right: a rendering of $\bar{z}$ after training, indicating that it encodes the shared dark background

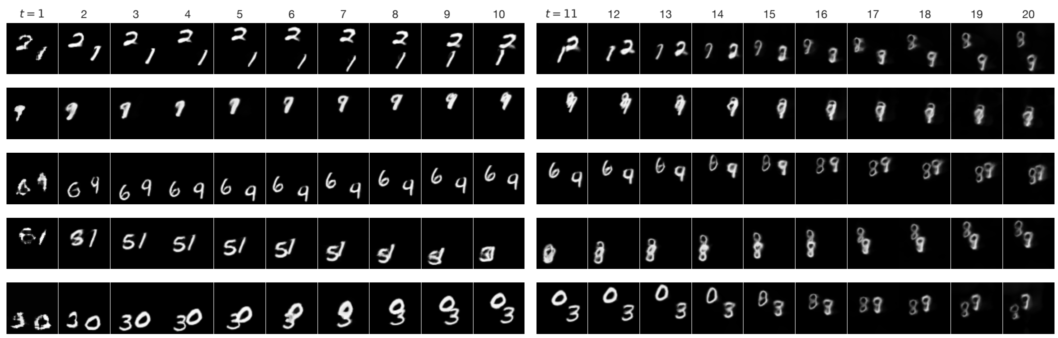


Figure 12: **NOVA's content identity is encoded in $\boldsymbol{z}_t$.** After $t = 10$ (context phase), we interpolate $\boldsymbol{z}_t \to \mathbf{0}$ while maintaining GCM-extracted actions $\boldsymbol{u}_t$. Digit identities gradually morph towards the base-network prototype (8s and 9s), while motion trajectories are preserved. No supervision on digit identity was provided.

D.2 Context-Conditioned Video Generation

D.2.1 Moving MNIST

Forecasting within the training horizon $T_{\text{inf}} = 20$. To complement results from Section 4.1, we generate a number of frames using Algorithm 1. Figure 14 shows qualitative video forecasting results on Moving MNIST, conditioned on 10 initial frames and rolling out for 10 predicted frames. NOVA produces sharp predictions without checkerboard artefacts, a common failure mode in transposed-CNN decoders.

Forecasting over longer horizons $T_{\text{inf}} = 1000$ Long-horizon generation is a potent test of identity consistency: a world model that merely interpolates over the temporal dimension will eventually collapse, while one with a structured representation should remain coherent far beyond its training horizon. We use Algorithm 1 to generate extremely long sequences. Given their storage and computational cost, we limit evaluation to two test sequences; we acknowledge this may not be fully representative of the test distribution.

Results in Figure 15 show that LAPO collapses, as its pixel distribution fully darkens and degenerates over time. Figure 16 and Table 7 further indicate that the standard WM is stable but for the wrong reason: it hedges toward a static, blurry distribution of pixels, hardly distinguishable from start to finish. NOVA, by contrast, commits. It picks digits and generates them, which means it can be wrong at times (visible as spikes in Figure 16), but the median metrics in Table 7 indicate that NOVA is more often right than wrong compared to the Standard WM.

D.2.2 PhyWorld

Unperturbed context-conditioned video generation. We also evaluate on PhyWorld under 32-step-long horizons, this time focusing on out-of-distribution (OOD) generalisation. Specifically, we test on ball radii and velocities that lie outside the training distribution, a regime where models without suitable inductive biases are expected to fail. Figure 17 shows four such sequences: in

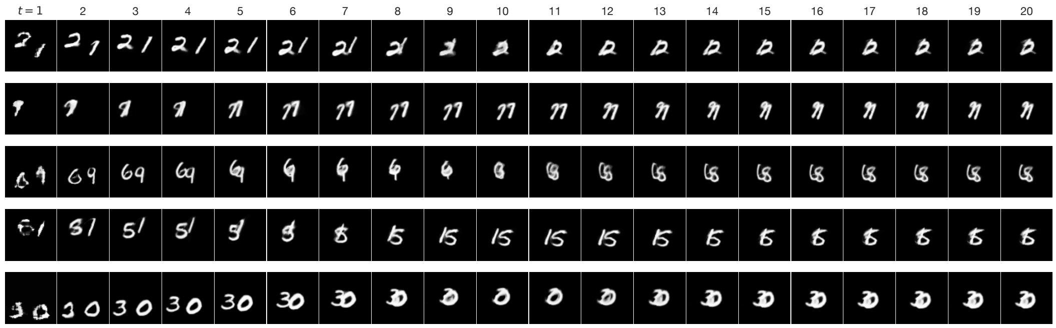


Figure 13: **NOVA's spatial motion is encoded in $\boldsymbol{u}_t$.** From $t = 0$, we interpolate $\boldsymbol{u}_t \to \mathbf{0}$ using GCM-generated actions, while keeping $\boldsymbol{z}_t$ free. Digits decelerate and converge towards the canvas centre, while their visual identities are fully preserved throughout.

Figure 14: **Context-conditioned video generation on Moving MNIST.** Frames 1–10 are context (ground truth); frames 11–20 are NOVA predictions. The model maintains digit identity and trajectory without artefacts.

each case, NOVA tracks the ground truth trajectory faithfully, correctly predicting both position and, implicitly, the physical properties of the balls. This suggests that the weight-space representation generalises beyond the training distribution, capturing the underlying collision dynamics rather than memorising seen configurations.

How far is video generation from world model [Kang et al., 2024]? Figure 18 shows three ground truth sequences and the resulting latent state interventions. We perform the intervention at step $t = 5$, well before the collisions happen. We see that once the new identity is inherited, post-collision velocities are different from the ones we would have expected from the ground truth. These adjusted collision dynamics appear visually accurate, suggesting that NOVA has effectively learned the physics of the PhyWorld environment, placing our latent state intervention experiment as a potent test of world modelling accuracy [Kang et al., 2024].

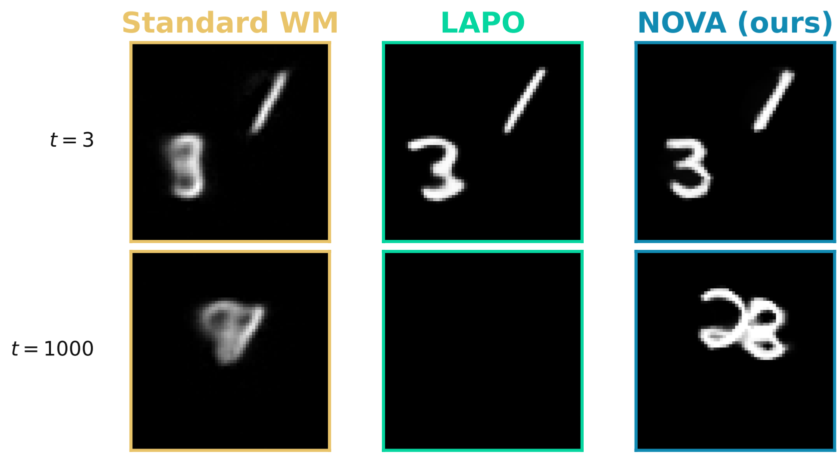


Figure 15: Comparison of long-horizon forecast ability. The models are only shown the initial two frames, and most autoregressively predict from frame $t = 3$ to $t = 1000$.

Table 7: Long-horizon identity-consistency metrics. We report the median, min, and max over $T = 1000$ frames and over two sequences $\{54, 57\}$, limited as such for computational reasons.

Metric	**Method**	**Median**	**Min**	**Max**
$W_1 \downarrow$	Standard WM	0.0780	0.0462	0.1308
	LAPO	0.0110	0.0037	0.0847
	NOVA	0.1222	0.0258	0.6138
JSD $\downarrow$	Standard WM	0.4255	0.3313	0.4966
	LAPO	0.0771	0.0703	0.2731
	NOVA	0.3204	0.1418	0.7994
SSIM $\uparrow$	Standard WM	0.5873	0.4337	0.8828
	LAPO	0.7683	0.4664	1.0000
	NOVA	0.5983	0.3552	0.9142
$d_{\mathcal{F}} \downarrow$	Standard WM	0.1224	0.0903	0.1982
	LAPO	0.0409	0.0234	0.2343
	NOVA	0.1953	0.1313	0.2517
$B \downarrow$	Standard WM	0.2698	0.1542	0.3785
	LAPO	0.0086	0.0071	0.1025
	NOVA	0.1442	0.0270	1.8013

Importantly, we take from Figure 18 that the actions were recalculated, confirming that latent actions and states are completely dissociated. Figure 18 shows that when actions are inherited, post-collision positions are the same (those of the alien sequence). These results indicate that physics are encoded in the action vector, and altering this vector would result in unphysical behaviour.

D.2.3 WeatherBench

We evaluate on WeatherBench with 24-step rollouts: 12 context frames followed by 12 predicted steps. Figure 20 shows that NOVA accurately reproduces large-scale weather fronts and avoids the excessive smoothing characteristic of phase 2 models trained with reconstruction MSE over the pixel space. We avoid this dilemma by computing the loss in the latent weight space.

Figure 16: **Long-horizon generation.** Comparison of long forecast ability using several metrics. The models are only shown the initial two frames, and most predict from frame $t = 3$ to $t = 1000$. This complements Figure 15 by adding time-dependent quantitative values.

D.3 Addressing Inference-Time Aliasing in INRs

Coordinate-based INRs map continuous coordinates to signals via high-frequency sinusoidal embeddings. When queried at a denser grid than the one used during training, as in super-resolution, frequency bands in the Fourier feature encoding may exceed the Nyquist limit of the training grid, producing aliasing artefacts [Barron et al., 2021; Lindell et al., 2022]. The effect is visible as horizontal bars in Figure 22, while Figure 21 illustrates why out-of-band frequencies are indistinguishable

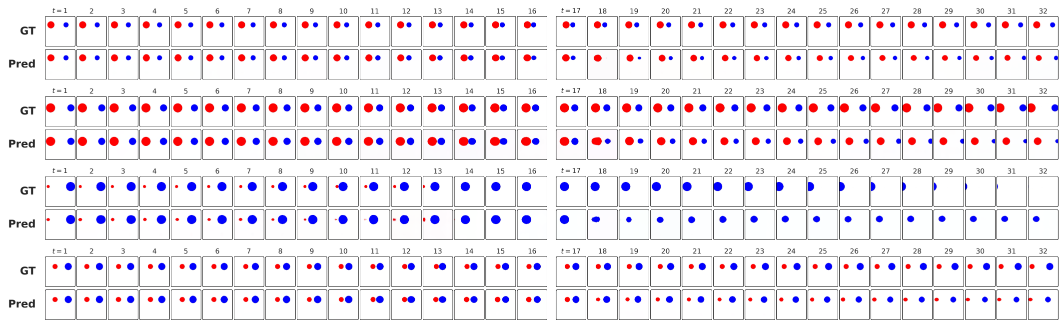


Figure 17: **OOD generalisation on PhyWorld.** Ground truth (GT) versus NOVA predictions for ball radii and velocities outside the training distribution.

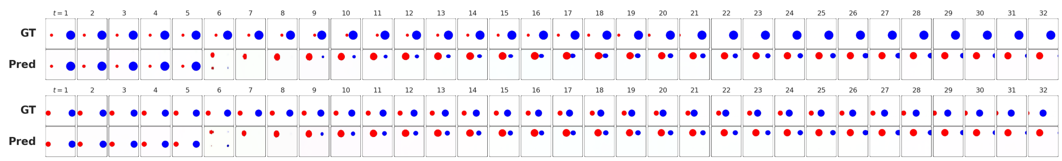


Figure 18: PhyWorld **state** intervention after 5 steps, with forecast generated using 3 context frames. Despite manually editing their identities, we observe accurate collision dynamics that respect the balls' sizes and velocities in the injected states. The fact that the y-position of the alien balls is inherited (in addition to shape and colour) is addressed in Section E.

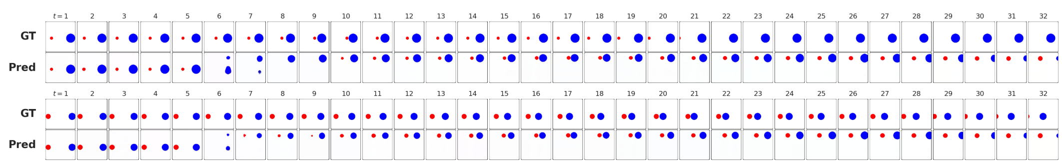


Figure 19: PhyWorld **action** intervention after 5 steps, with forecast generated using 3 context frames. We use the same sequences as in Figure 18. Injected alien collision dynamics (actions) are the same, leading to unphysical behaviours.

from in-band ones at the training sample locations.[12] The loss cannot differentiate them, so the corresponding weights are optimised against noise and produce artefacts at super-resolution.

Theorem 1 (Nyquist–Shannon Sampling Theorem (Informal))**.** *If a continuous function $f(y)$ is sampled at uniform intervals Δy, it can be unambiguously reconstructed if and only if f contains no spatial frequencies exceeding $\xi_{\text{Nyquist}} = \frac{1}{2\Delta y}$.*

For a grid of H pixels over $y \in [-1, 1]$, the sampling interval is $\Delta y = 2/H$, giving $\xi_{\text{Nyquist}} = H/4$ periods per unit length. The standard Fourier feature encoding assigns frequency $\xi_k = 2^{k-1}$ to band k as in [Tancik et al., 2020]

$$\gamma(y)_k = \big(\sin(2^k \pi y),\ \cos(2^k \pi y)\big), \qquad k \in \{0, 1, \ldots, K-1\}, \tag{7}$$

or in full,

$$\gamma(y) = \big(\sin(2^0 \pi y),\, \cos(2^0 \pi y),\ \ldots,\ \sin(2^{K-1} \pi y),\, \cos(2^{K-1} \pi y)\big)^\top \quad \in \mathbb{R}^{2K}. \tag{8}$$

The no-aliasing condition $\xi_k < \xi_{\text{Nyquist}}$ requires:

$$2^{k-1} < \frac{H}{4} \implies k_{\max} < \log_2(H) - 1. \tag{9}$$

For MiniGrid ($H = 72$), $k_{\max} < 5.17$; therefore, all $K = 6$ bands remain within the safe manifold. However, for WeatherBench ($H = 32$), $k_{\max} < 4$, meaning that the highest-frequency bands must be eliminated.

[12] For our analysis, we consider aliasing along a single axis y for clarity, as the two spatial dimensions are separable.

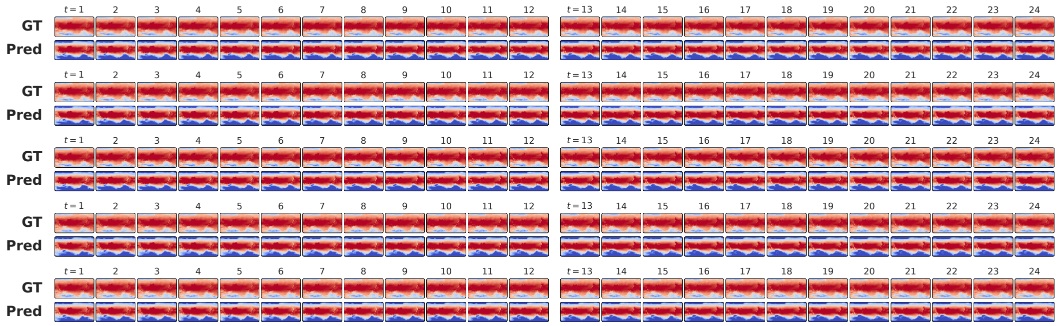


Figure 20: **WeatherBench forecasting.** Ground truth (GT) versus NOVA predictions conditioned on 12 initial frames (frames 1–12) and rolling out for 12 steps (frames 13–24).

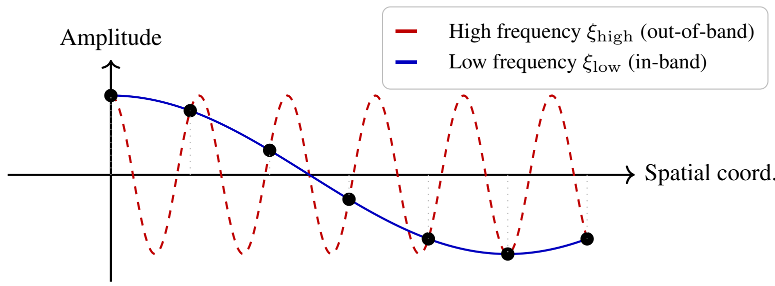


Figure 21: **Illustration of spatial aliasing.** The out-of-band signal (red, dashed) coincides with the in-band signal (blue) at every training sample point (black dots). During training, however, the loss cannot distinguish the two; weights are optimised against noise, producing artefacts when the INR is queried at a finer out-of-band grid.

To suppress these, we remark that entries $j = 2k$ and $j = 2k+1$ in $\gamma(y)$ correspond to band k. We define a binary mask $\mathbf{m} \in \{0,1\}^{2K}$ with

$$m_{2k} = m_{2k+1} = \mathbf{1}[k < \log_2(H) - 1], \qquad k = 0, \dots, K-1, \tag{10}$$

and evaluate queries using the masked embedding $\tilde{\gamma}(y) = \mathbf{m} \odot \gamma(y)$, where $\odot$ denotes element-wise multiplication. This zeros out every frequency band that violates Equation (9), projecting onto the aliasing-free subspace of $\mathbb{R}^{2K}$ induced by the training grid. A similar process is independently applied to the x coordinate.

Bespoke architectures such as BACON [Lindell et al., 2022] address this by enforcing band-limiting through specialised multiplicative filter networks at training time. Our principled approach, directly inspired by the Nyquist–Shannon Sampling Theorem (see Theorem 1), requires only a single masking matrix applied to the Fourier embedding at inference time, eliminating aliasing zero-shot, with no architectural changes or retraining.

D.4 Navigating Discrete Action Spaces within MiniGrid

For domains with discrete actions such as **MiniGrid** [Chevalier-Boisvert et al., 2023], we design a discrete NOVA variant. We employ codebooks [Van Den Oord et al., 2017], and we quantise the continuous prediction $\boldsymbol{u}_t$ to its nearest codebook vector $\tilde{\boldsymbol{u}}_t$. To enable backpropagation through this non-differentiable operation, we apply the straight-through estimator (STE), redefining the action as $\tilde{\boldsymbol{u}}_t = \boldsymbol{u}_t + \mathtt{sg}(\tilde{\boldsymbol{u}}_t - \boldsymbol{u}_t)$. The FDM is then evaluated on $\tilde{\boldsymbol{u}}_t$, and the objective (2) is augmented with the standard codebook and commitment losses to align the embeddings.

Figure 24 shows NOVA's behaviour cloning GCM operating on MiniGrid. Although the environment has a small set of canonical discrete actions (turn left, turn right, move forward, toggle), we employed 200 actions, hoping to form action clusters as done by Schmidt and Jiang [2023]. Figure 23 (left) shows soft clusters around the quantised actions formed by the raw GCM-predicted actions.

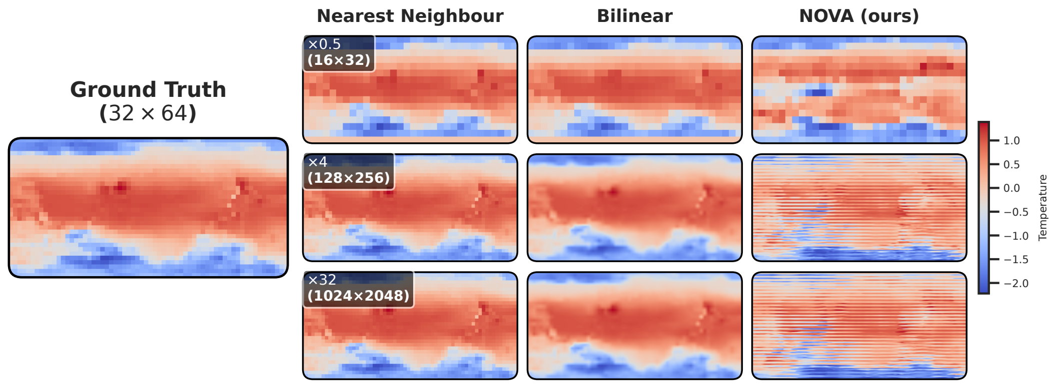


Figure 22: **Zero-shot super-resolution on WeatherBench without frequency masking.** Compared to Figure 7, horizontal bars are visible, indicating spectral aliasing along the vertical axis.

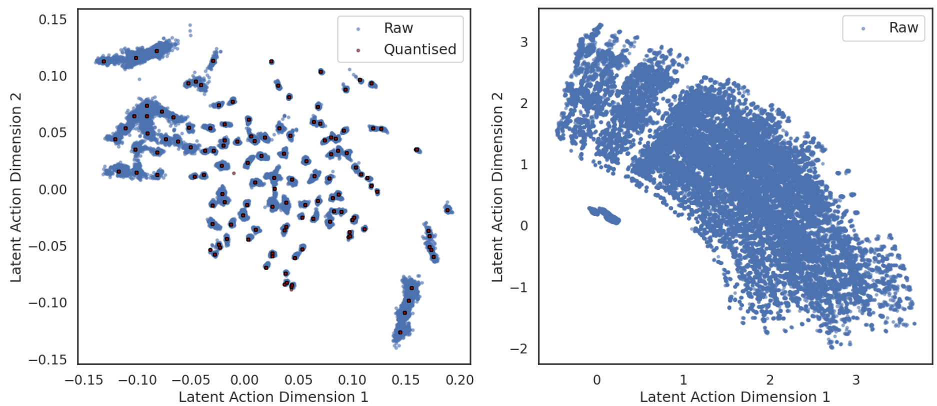


Figure 23: **Latent action dimensions on MiniGrid. Discrete** (Left): Raw (grey) and quantised (blue). **Continuous** (Right) The first two dimensions collectively encode something akin to the orientation of the agent, while the other two (not shown here) encode its location.

We also implement the continuous variant as defined in Figure 2 for direct comparison. Continuous actions train faster and drive the loss roughly two orders of magnitude lower, yielding better visual quality in the downstream GCM-based behaviour cloning rollouts (seen by comparing Figure 25 to Figure 24). Furthermore, the continuous action geometry appears more interpretable, as Figure 23 implies that latent action dimensions 1 and 2 encode the orientation of the agent (red triangle). Taken together, these results suggest that the NOVA framework is more naturally suited to continuous action representations. This is consistent with the finding of Garrido et al. [2026] that discrete codebooks are insufficient for unlabelled videos in the wild.

D.5 Comparison to Weight-Space Linear Recurrent Neural Networks

Within the emerging literature on Weight-Space Learning (WSL), our proposed framework (specifically, NOVA's GCM in phase 3) shares several similarities with contemporary architectures, most notably Weight-space Adaptive Recurrent Prediction (WARP) [Nzoyem et al., 2026]. Both frameworks view the latent state of the dynamic system not as a standard feature vector, but as the weights and biases of an INR. In WARP, this is governed by a continuous linear recurrence defined as $\boldsymbol{z}_t = A\boldsymbol{z}_{t-1} + B\Delta\boldsymbol{o}_t$, where the input difference $\Delta\boldsymbol{o}_t$ between consecutive frames acts as an implicit action driving the process. Despite these foundational similarities, WARP is a general sequence model, whereas our framework is designed as a comprehensive world model for modelling complex video dynamics.

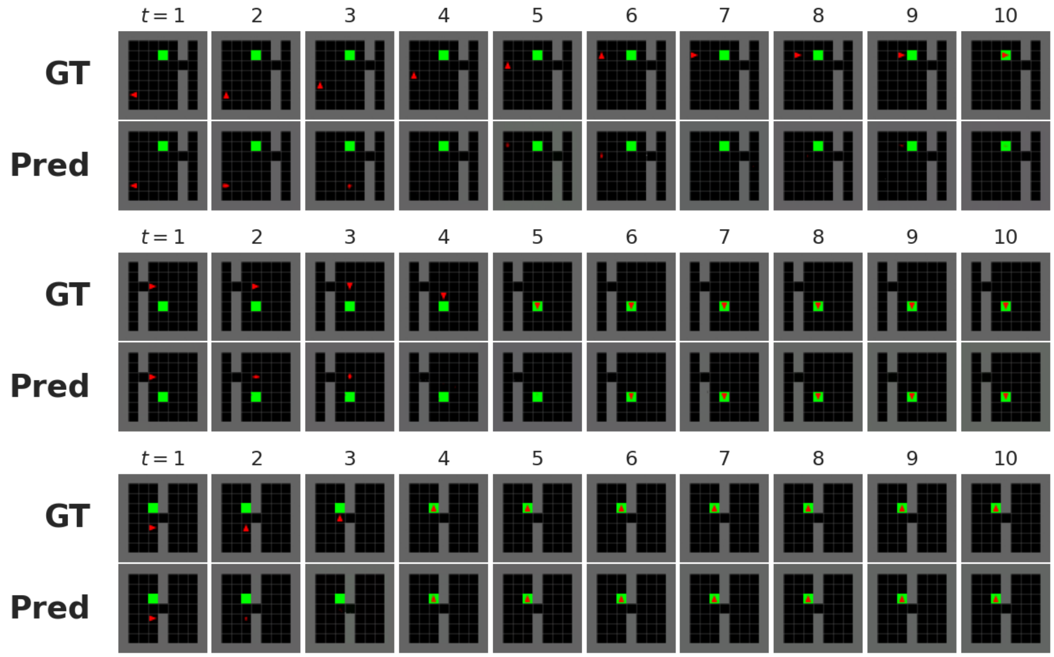


Figure 24: Goal-directed navigation within a ***discrete*** action space. The GCM, conditioned only on the initial frame, imitates the BFS policy to navigate towards the goal (green box). Ground truth (top row) vs. NOVA prediction (bottom row) per sequence.

Figure 25: Goal-directed navigation within a ***continuous*** action space. Similar to Figure 24, the agent (red triangle) is driven by GCM-generated actions.

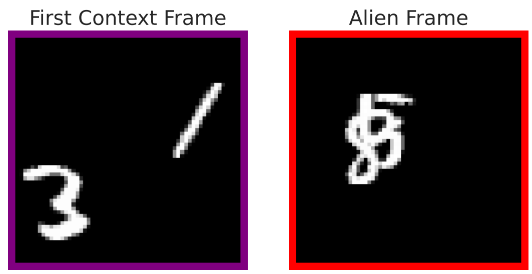
Implementation To conduct an empirical comparison, we implemented the WARP architecture utilising the JAX [Bradbury et al., 2018] and Equinox [Kidger and Garcia, 2021] libraries. We meticulously controlled the experimental environment to ensure parity with our framework's training conditions, matching the batch size and total computational budget. The WARP baseline was trained using a fixed learning rate of 10^{-6}. Given stability issues [Nzoyem et al., 2026, p. 23], we stabilise the linear recurrence and prevent the unbounded explosion by enforcing weight clipping $\boldsymbol{z}_t \in [-0.5, 0.5]$. Furthermore, the decoded outputs were subjected to pixel clipping between 0 and 1. We matched the capacity of NOVA by reusing an identical INR of dimension $d_{\boldsymbol{z}} = 961$. Finally, the same NOVA CNN encoder architecture was utilised to act as WARP's initial hypernetwork ϕ, mapping the initial visual observation into the high-dimensional weight space: $x_0 \mapsto \boldsymbol{z}_0$.

Forecasting results. Standard evaluation of the forecasted rollouts (Figure 26) demonstrates that WARP is somewhat capable of capturing spatial-temporal dynamics when conditioned on 10 previous context frames. However, qualitative assessments of the generated sequences reveal a substantial degradation in visual fidelity. The predicted digits suffer from severe artifacting; they are noticeably blurry, ghosted, and lack crisp, well-defined boundaries. This visual degradation suggests that WARP's unconstrained linear recurrence struggles to balance two competing objectives: maintaining accurate physical trajectories and achieving photorealism.

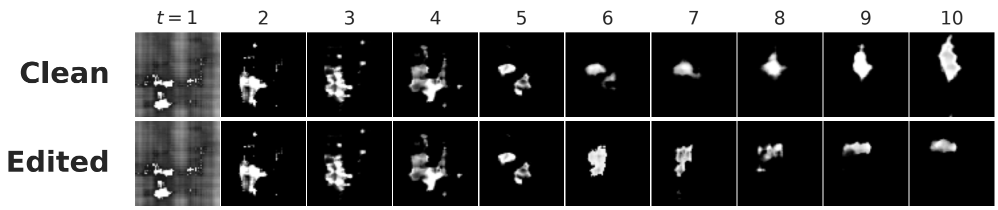
Figure 26: WARP forecasting results conditioned on $t = 10$ context frames. While the model tracks general temporal dynamics, it fails to maintain sharp visual fidelity, yielding blurred and ghosted predictions.

Latent disentanglement analysis. To probe the underlying topology of WARP's learned representations, we performed the latent state intervention experiment from Figure 1. Following the processing of the initial 3 frames and autoregressive generation after those, we intervened at $t = 6$ by replacing the naturally evolved state $\boldsymbol{z}_t$ with the latent encoding of an alien frame.

However, the intervention sequence (Figure 27b) demonstrates a catastrophic failure, as WARP allows the alien injection to fundamentally corrupt the forward dynamics (similar to LAPO). Furthermore, Figure 27a indicates that the CNN encoder appears unable to reconstruct the latents with high visual quality even in isolation. This is likely because the initial representation $\boldsymbol{z}_0$ it is forced to generate must be mathematically compatible with the subsequent linear recurrence. We believe this imposes an architectural compromise, yielding an initial state that neither perfectly suits the unrolled recurrence nor permits faithful visual reconstruction. This behaviour proves that the latent representations learned by WARP are thoroughly entangled, with spatial content, temporal coordinates, and motion confounded within z_t.



(a) Initial context frame and ground truth alien frame for latent intervention.



(b) Intervention sequence rollout

Figure 27: Latent State Intervention in WARP. (a) The original context frame alongside the isolated decoding of the alien latent state. Note the poor baseline reconstruction quality of the encoder. (b) When the alien state is substituted at $t = 6$, the linear recurrence is fundamentally disrupted, proving that content and motion are highly entangled within z_t.

D.6 Additional Ablation Studies

Throughout our study, we performed several ablation studies, such as additive vs monolithic FDM Section 4.2, and discrete vs continuous action spaces in Section D.4. We use this section to present additional abstractions that justify our architectural choices.

Removing the $\bar{z}$ anchor. Ablating the base parameter $\bar{z}$ and predicting full INR weights z_t instead of relative offsets fundamentally destabilises the learning process, trapping the network in local minima for prolonged periods (see Figure 28). This observation suggests that $\bar{z}$ prevents severe

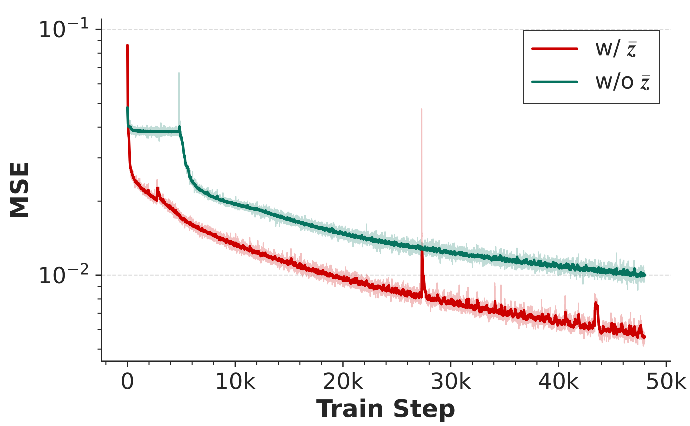


Figure 28: Phase 2 training loss with and without $\bar{z}$ on Moving MNIST.

bottlenecks and gradient instabilities that arise when the model is forced to independently regress the full high-dimensional weight space from scratch at every time step.

Replacing the recurrent GCM architectures. In our study, we observed on PhyWorld 30K that latent interventions are most effective when paired with non-attention-based GCM architectures such as recurrent neural networks. We hypothesise that this is because standard Transformers compute global attention over the entire state-action history. As a result, injecting an alien action mid-sequence causes attention disruption and visual artifacting.

E Limitations & Failure Modes

Spurious information encoded within latent states and actions. On Moving MNIST, the latent action $\boldsymbol{u}_t$ encodes not only the next digit location but also brightness variations. This is partly because the IDM is trained to explain *all* inter-frame differences, including those due to noise [Zhang et al., 2025a]. Furthermore, because the actions encode next digit location "go to x, y" instead of saying "increment by $\Delta x, \Delta y$", it becomes harder to map actions to ground truths when available [Schmidt and Jiang, 2023]. On PhyWorld, we find that intervening on the content also shifts the movement along the vertical (see Figure 29); we hypothesise that this is because during training, the model only sees balls on the same axis, and thus encodes the y-location as part of the identity. Future work could explore regularisation mechanisms beyond discretisation [Garrido et al., 2026] to constrain the latent spaces. For example, applying SICReg [Maes et al., 2026] to bottleneck the action space and encode only meaningful user-mappable motion information (for Moving MNIST and MiniGrid), and data augmentations such as video rotations for physics-abiding datasets such as PhyWorld.

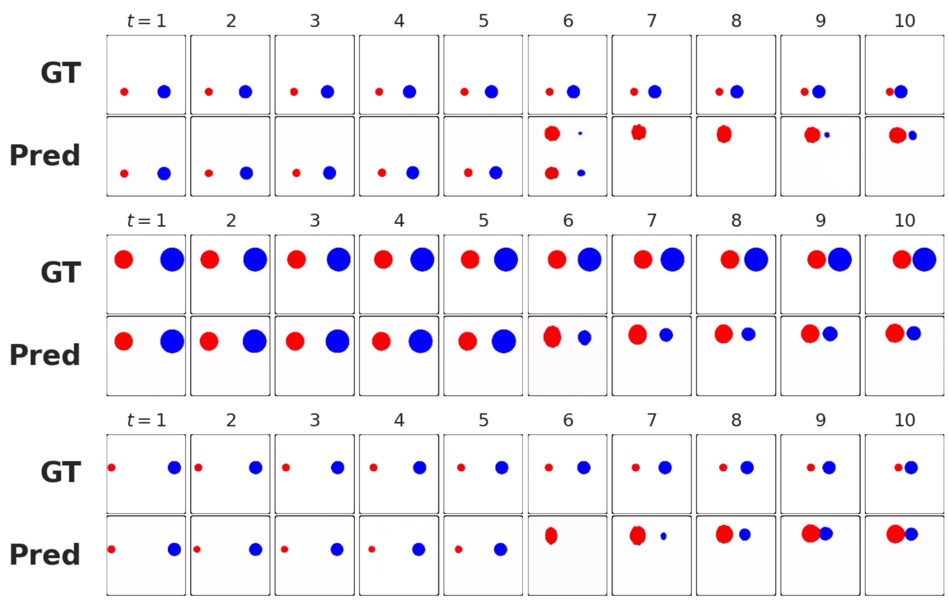


Figure 29: Zero-shot content retargeting on PhyWorld. Intervention starts at $t = 5$, with all subsequent states replaced with those corresponding to the same alien sequence. While all sequences inherit new ball colours and shapes, we find that in addition, the y-position is inherited, thus altering visual dynamics, which should be free. Future work will aim to make the model robust to such spurious correlations while preserving dynamics.

Physical constraints and OOD generalisation. On PhyWorld, we observed diminished OOD performance after suppressing the addition and using a single monolithic FDM (see Figure 30 and Table 8). We believe scientific machine learning techniques [Rackauckas et al., 2020] could help alleviate this issue, whereby the INR would be forced to satisfy laws of motion such as $\boldsymbol{z}_{t+1} = \boldsymbol{z}_t + \Delta t \cdot \boldsymbol{u}_t$ [Nzoyem et al., 2026], or any other partial differential equation thanks to the implicit function theorem [Bai et al., 2019].

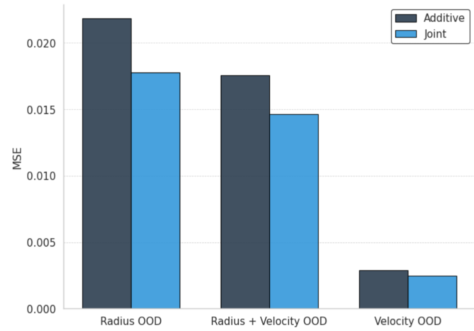


Figure 30: Additive vs. monolithic (joint) forward dynamics on PhyWorld. Although detrimental for disentangled representations, a joint FDM performs better out-of-distribution.

Table 8: Comparison of additive and joint model performance on in-distribution (InD) and out-of-distribution (OOD) data after phase 2 training (i.e., using the IDM to generate actions). Values are reported as mean $\pm$ standard deviation.

Distribution	**Model**	**Image Metrics**			**Physical Metrics**		
		MSE $\downarrow$	**PSNR** $\uparrow$	**SSIM** $\uparrow$	**Pos. Error** $\downarrow$	**Mom. Error** $\downarrow$	**K.E. Error** $\downarrow$
InD	Additive	0.0008 ± 0.0004	32.52 ± 1.52	0.9792 ± 0.0072	0.0065 ± 0.0188	0.0308 ± 0.0455	0.0404 ± 0.2024
	Joint	0.0007 ± 0.0003	32.64 ± 1.40	0.9803 ± 0.0071	0.0031 ± 0.0051	0.0191 ± 0.0099	0.0041 ± 0.0049
OOD	Additive	0.0141 ± 0.0194	24.70 ± 6.23	0.9255 ± 0.0701	0.0739 ± 0.0883	0.4109 ± 0.8788	0.6531 ± 1.8469
	Joint	0.0117 ± 0.0160	25.08 ± 5.87	0.9297 ± 0.0649	0.0753 ± 0.0882	0.3380 ± 0.6255	0.6463 ± 1.7010

SOTA video generation. While our framework enables controllable generation, we do not achieve state-of-the-art context-conditioned video forecasting. Supervised spatio-temporal models from the OpenSTL benchmark [Tan et al., 2023] achieve highly optimised pixel-level metrics. This performance gap arises from an architectural constraint that forces NOVA to autoregressively rely on an inferred latent action signal to drive the generation process. Similar trade-offs where world models sacrifice raw pixel-level reconstruction performance to prioritise powerful latent representations have recently been proposed [Terver et al., 2026]. That said, bridging this gap remains an exciting frontier for future work, paving the way for architectures that seamlessly unite state-of-the-art visual fidelity with interpretable, action-driven control.